Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

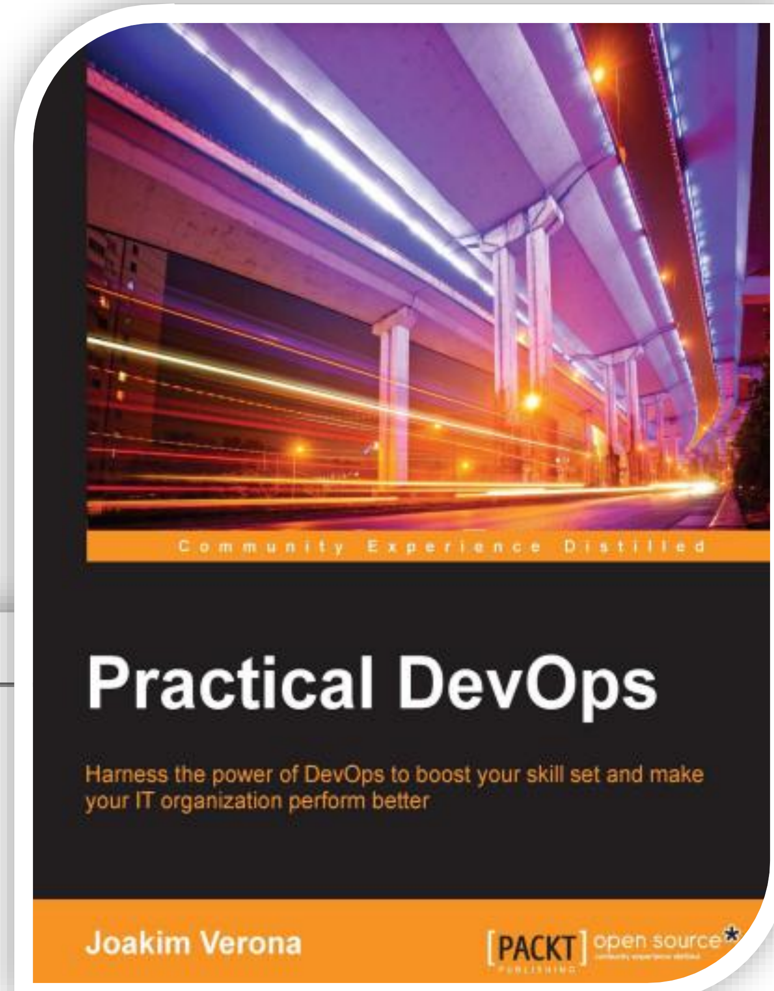
Issue Tracking

Security & database management *in* cloud server

Welcome to *the* Development Operation Engineering *course*

Chapter 9:
Issue Tracking

Chapter 9: Issue Tracking



What are issue trackers used for?

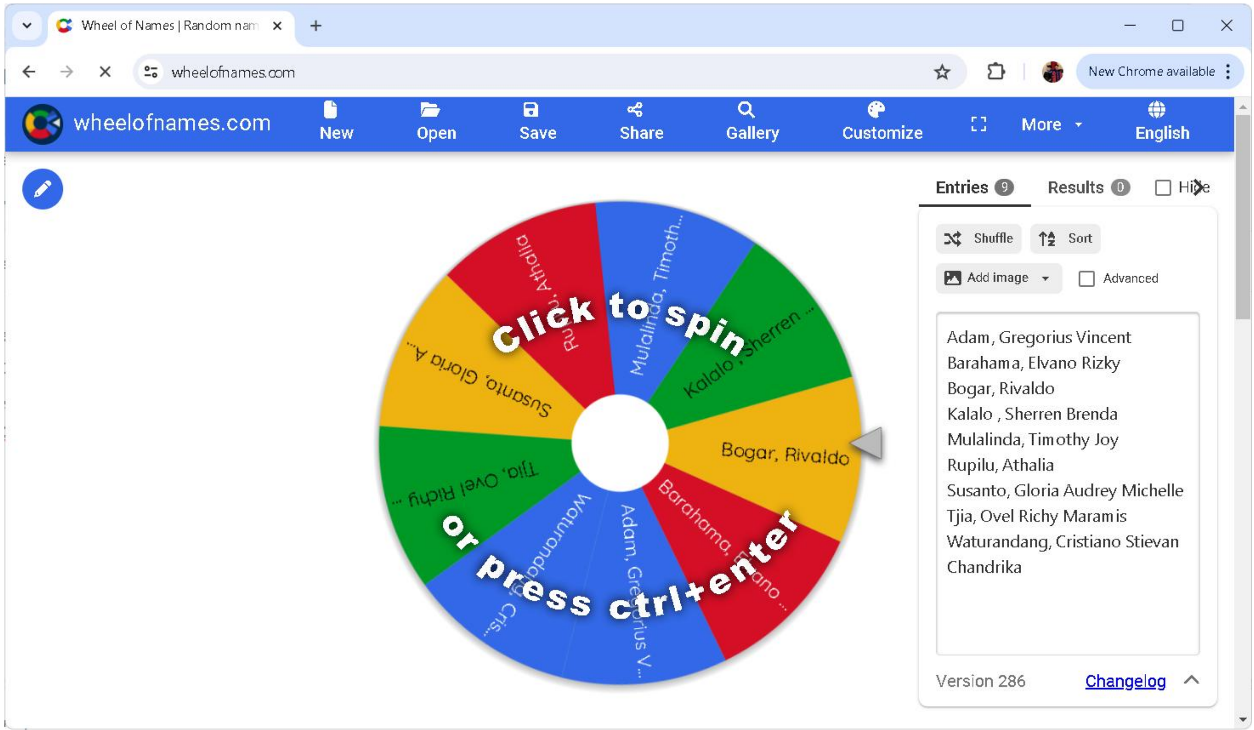
What are issue trackers used for?

From an Agile process standpoint, issue trackers are used to help with the minutiae and details of the Agile process. The entities handled by the issue tracker might represent work items, bugs, and issues. Most Agile processes include an idea of how to manage tasks that an Agile team are to perform, in the form of Post-it notes on a board or an electronic equivalent.

When working in an Agile setting, it is common to have a board with issues on handwritten Post-it notes. This is a central concept in the Kanban method, since Kanban actually means *signboard* in Japanese. The board gives a nice overview of the work in progress and is pretty easy to manage because you just move the Post-it notes around on the board to represent state changes in the workflow.

Issue Tracking allows teams to track issues or bugs that occur in the software development process. By monitoring and logging every issue that arises, teams can identify **patterns**, **trends**, or recurring **problems** (*masalah berulang*) that may require special attention (*perhatian khusus*).


ERROR
ITU BUKAN
MUSUH
Belajarlah dari setiap kesalahan!
404
ERROR
ERROR



Random Picker:

https://wheelofnames.com/

Bug tracking di DevOps dan SDLC bertujuan untuk mengelola dan melacak bugs atau isu-isu yang ditemukan dalam pengembangan system/software. Apakah sama *bug tracking* di DevOps dan SDLC?

Electronic Issue Tracker

When project managers first encounter an electronic issue tracker, there is often a tendency to go completely overboard with adding complexity to the state machines and attribute storage of the tracker. That said, there is clearly a lot of useful information that can be added to an issue tracker without overdoing it.

Apart from the previously mentioned basic information, these additional attributes are usually quite useful:

- **Due date**: The date on which the issue is expected to be resolved.
- **Milestone**: A milestone is a way to group issues together in useful work packages that are larger than a single issue. A milestone can represent the output from a Scrum sprint, for example. A milestone usually also has a due date, and if you use the milestone feature, the due dates on individual issues are normally not used.
- **Attachments**: It might be convenient to be able to attach screenshots and documents to an issue, which might be useful for the developer working on the issue or the tester verifying it.
- **Work estimates**: It can be useful to have an estimate of the expected work expenditure required to resolve the issue. This can be used for planning purposes. Likewise, a field with the actual time spent on the issue can also be useful for various calculations. On the other hand, it is always tricky to do estimates, and in the end, it might be more trouble than it's worth. Many teams do quite well without this type of information in their issue tracker.

Agile Workflow Model

The following are also some useful states that can be used to model an Agile workflow better than the plain open and closed states. The open and closed states are repeated here for clarity:

- **Open**: The issue is reported, and no one is working on it as of yet
- **In progress**: Somebody is assigned to the issue and working on resolving it
- **Ready for test**: The issue is completed and is now ready for verification. It is unassigned again
- **Testing**: Someone is assigned to work on testing the implementation of the issue
- **Done**: The task is marked as ready and unassigned once more. The done state is used to mark issues to keep track of them until the end of a sprint, for example, if the team is working with the Scrum method
- **Closed**: The issue is no longer monitored, but it is still kept around for reference

Why issue tracker?

What do we need from an issue tracker software?

- What scale do we need?

  Most tools work well on scales of up to about 20 people, but beyond that, we need to consider performance and licensing requirements. How many issues do we need to be able to track? How many users need access to the issue tracker? These are some of the questions we might have.

- How many licenses do we need?

  In this regard, free software gains the upper hand, because proprietary software can have unintuitive pricing. Free software can be free of charge, with optional support licensing.

  Most of the issue trackers mentioned in this chapter are free software, with the exception of Jira.

- Are there performance limitations?

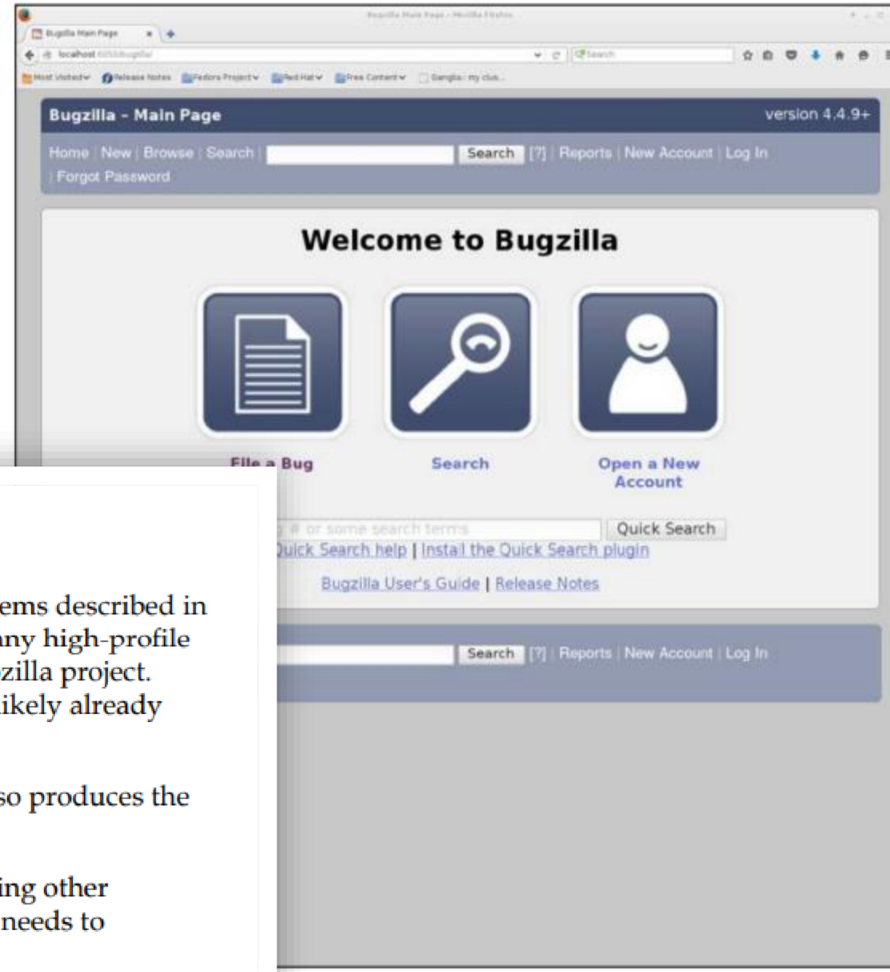
  Performance is not usually a limiting factor, since most trackers use a production-ready database such as PostgreSQL or MariaDB as the backend database. Most issue trackers described in this chapter behave well at the scales usually associated with an organization's in-house issue tracker. Bugzilla has been proven in installations that handle a large number of issues and face the public Internet.

  Nevertheless, it is good practice to evaluate the performance of the systems you intend to deploy. Perhaps some peculiarity of your intended use triggers some performance issue. An example might be that most issue trackers use a relational database as a backend, and relational databases are not the best for representing hierarchical tree structures if one needs to do recursive queries. In normal use cases, this is not a problem, but if you intend to use deeply nested trees of issues on a large scale, problems might surface.

Trackers allow teams to track the progress of a particular issue or task.

By monitoring the status and progress of each issue or change, teams can ensure that nothing is forgotten or left behind.

Issue Tracker Systems



Bugzilla

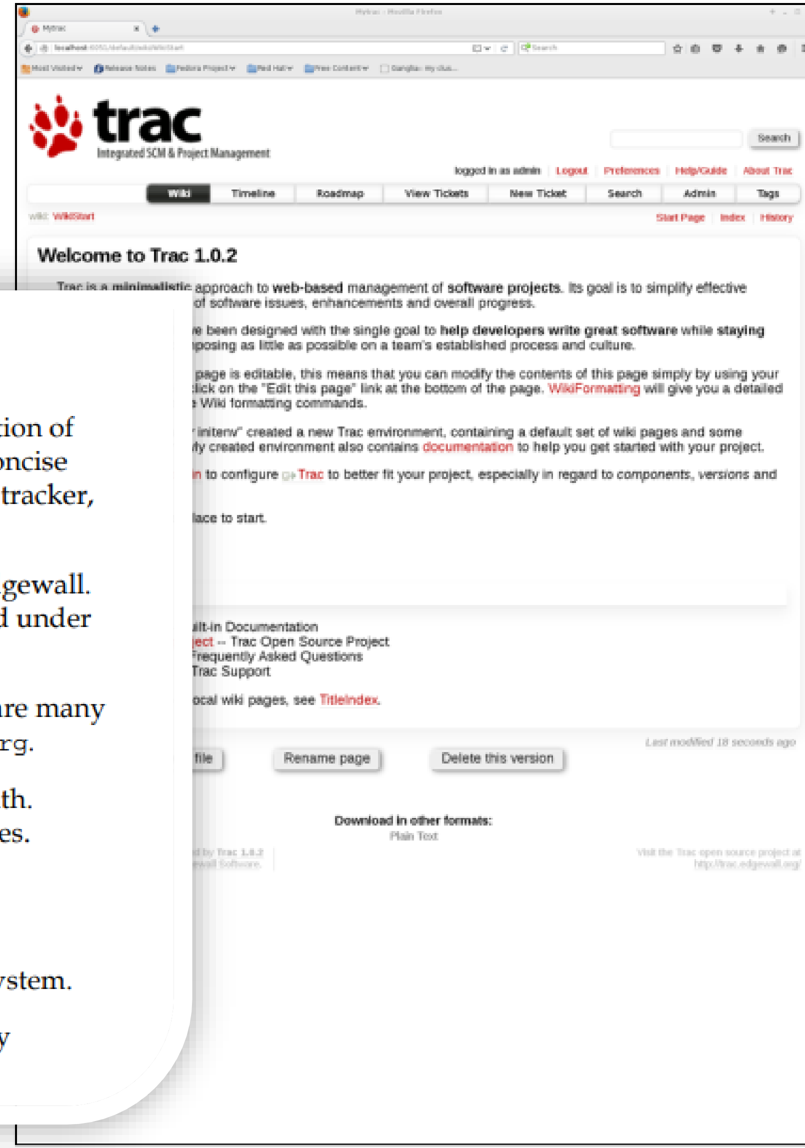
Bugzilla can be said to be the grandfather of all the issue tracker systems described in this chapter. Bugzilla has been around since 1998 and is in use by many high-profile organizations, such as Red Hat, the Linux kernel project, and the Mozilla project. If you have ever reported a bug for one of these products, you have likely already encountered Bugzilla.

Bugzilla is free software maintained by the Mozilla project, which also produces the well-known Firefox browser. Bugzilla is written in Perl.

Bugzilla focuses, unsurprisingly, on tracking bugs rather than handling other issue types. Handling other types of tasks, such as providing a wiki, needs to be done separately.

Many organizations use Bugzilla as a public-facing tool where issues can be reported. As a result, Bugzilla has good security features.

Trac Issue Tracker



Trac

Trac is an issue tracker that is fairly easy to set up and test. The main attraction of Trac is that it is small and integrates a number of systems together in one concise model. Trac was one of the earliest issue trackers that showed that an issue tracker, a wiki, and a repository viewer could be beneficially integrated together.

Trac is written in Python and published under a free software license by Edgewall. It was originally licensed under the GPL, but since 2005, it has been licensed under a BSD-style license.

Trac is highly configurable through the use of a plugin architecture. There are many plugins available, a number of which are listed on `https://trac-hacks.org`.

Trac develops at a conservative pace, which many users are quite happy with. Trac also has an XML-RPC API, which can be used to read and modify issues.

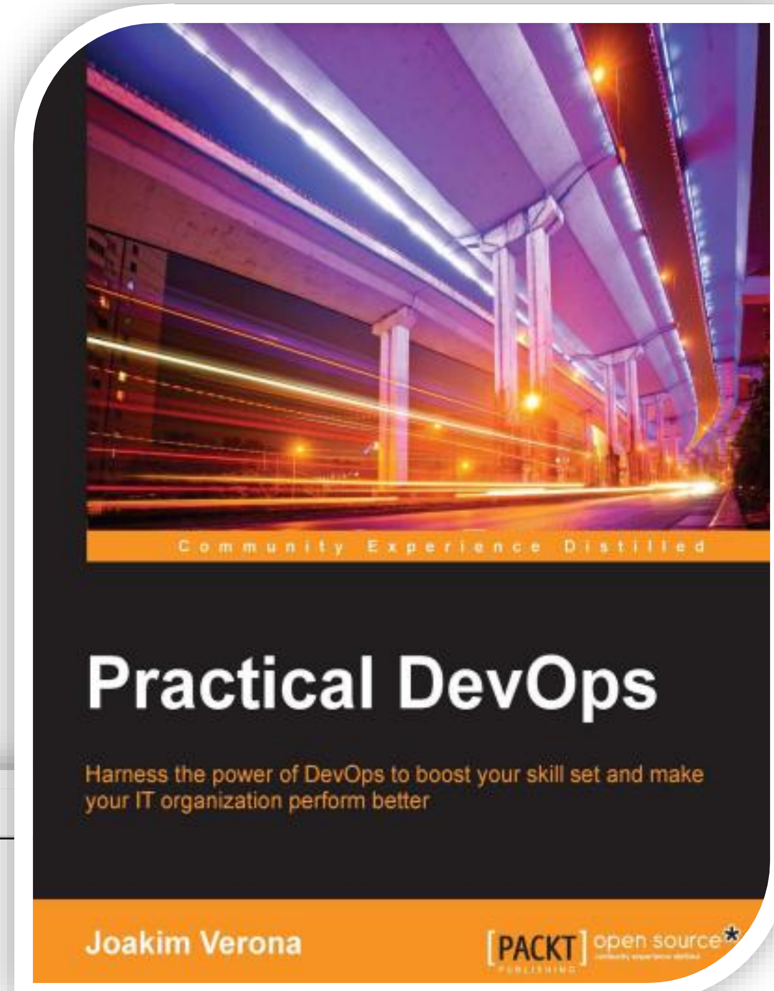
Trac traditionally used to integrate only with Subversion, but now features Git support.

Trac also has an integrated wiki feature, which makes it a fairly complete system.

There are several Docker hub Trac images available, and Trac is also usually available in the repositories of Linux distributions.

Chapter 10:

The Internet of Things and DevOps



Community Experience Distilled
Practical DevOps
Harness the power of DevOps to boost your skill set and make your IT organization perform better
Joakim Verona
[PACKT] open source

Types *of* IoT Devices

How can DevOps assist us in the emerging field of the Internet of Things? The Internet of Things, or IoT for short, presents challenges for DevOps.

- **Smartwatches**: Today, a smartwatch can have a Bluetooth and Wi-Fi connection and automatically detect available upgrades. It can download new firmware and upgrade itself on user interact
smartwatches available today, from the robust
and Apple Watch. There already are smartwa
connections as well, and eventually, they will
- **Keyboards**: A keyboard can have upgradeabl
actually a good example of an IoT device, in a
It can run software that provides sensor reado
that in the end are Internet-connected. There a
keyboards with open firmware, such as the Er
Arduino-style board to a keyboard matrix (ht
- **Home automation systems**: These are Interne
controlled with smartphones or desktop comp
an example of such a device that lets you cont
Internet-connected device. There are many ot
- **Wireless surveillance cameras**: These can be
interface. There are many providers for differ
Axis Communications.

- **Biometric sensors**: Such sensors include fitness sensors, for example, pulse meters, body scales, and accelerometers placed on the body. The Withings body scale measures biometrics and uploads it to a server and allows you to read statistics through a website. There are accelerometers in smartwatches and phones as well that keep track of your movements.
- **Bluetooth-based key finders**: You can activate these from your smartphone if you lose your keys.
- **Car computers**: These do everything from handling media and navigation to management of physical control systems in the car, such as locks, windows, and doors.
- **Audio and video systems**: These include entertainment systems, such as networked audio players, and video streaming hardware, such as Google Chromecast and Apple TV.
- **Smartphones**: The ubiquitous smartphones are really small computers with 3G or 4G modems and Wi-Fi that connects them to the wider Internet.
- **Wi-Fi capable computers embedded in memory cards**: These make it possible to convert an existing DSLR camera into a Wi-Fi capable camera that can automatically upload images to a server.
- **Network routers in homes or offices**: These are in fact, small servers that often can be upgraded remotely from the ISP side.
- **Networked printers**: These are pretty intelligent these days and can interact with cloud services for easier printing from many different devices.

IoT Deployment Affects Software Architecture

An IoT network consists of many devices, which might not be at the same firmware revision. Upgrades might be spread out in time because the hardware might not be physically available and so on. This makes compatibility at the interface level important. Since small networked sensors might be memory and processor constrained, versioned binary protocols or simple REST protocols may be preferred. Versioned protocols are also useful in order to allow things with different hardware revisions to communicate at different versioned end points.

Massive sensor deployments can benefit from less talkative protocols and layered message queuing architectures to handle events asynchronously.

IoT deployment security

Security is a difficult subject, and having lots of devices that are Internet-connected rather that on a private network does not make the situation easier. Many consumer hardware devices, such as routers, have interfaces that are intended to be used for upgrades but are also easy to exploit for crackers. A legitimate service facility thus becomes a backdoor. Increasing the available surface increases the number of potential attack vectors.

DevOps & IoT, it is possible?

- ❖ We have discussed the basics of the Internet of Things, which is basically our ordinary Internet but with many more nodes than what we might normally consider possible.
- ❖ We have also seen that in the next couple of years, the number of devices with Internet capability in some form or another will keep on growing exponentially. Much of this growth will be in the **machine-to-machine parts of the Internet**.
- ❖ But is **DevOps, with its focus on fast deliveries**, **really the right fit for large networks of critical embedded devices?**
- ❖ The classic counterexamples would be DevOps in a nuclear facility or in medical equipment such as pacemakers. But just making faster releases isn't the core idea of DevOps. It's to make faster, more correct releases by bringing people working with different disciplines closer together.
- ❖ This means bringing production-like testing environments closer to the developers and the people working with them closer together as well. Described like this, it really does seem like DevOps can be of use for traditionally conservative industries.

DevOps and IoT (Internet of Things) can definitely be integrated. In fact, combining DevOps practices with IoT development can bring numerous benefits, especially in terms of managing complexity, ensuring reliability, and accelerating time-to-market for IoT solutions.

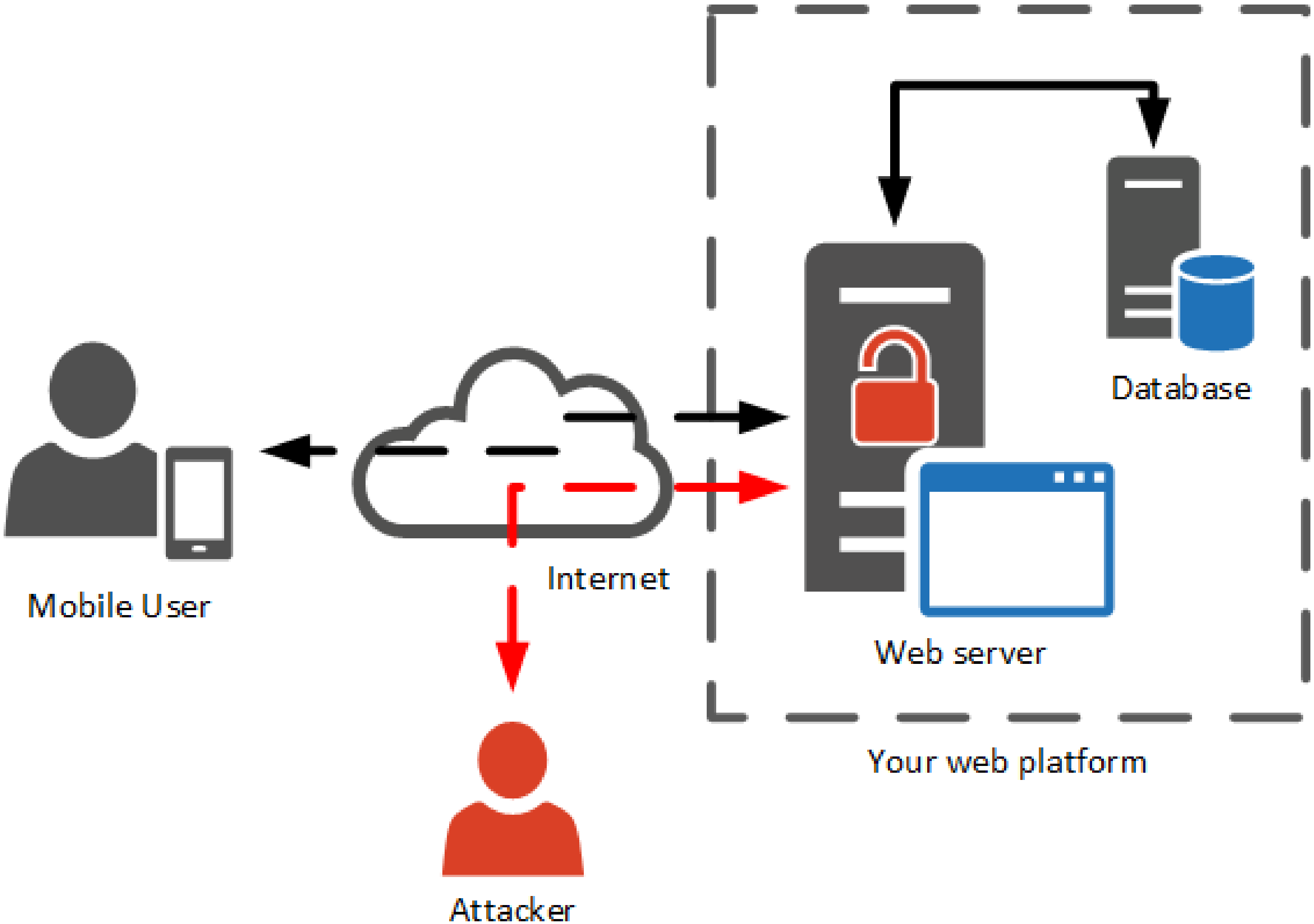
Dalam implementasi dari DevOps, kira-kira mana yang paling sulit, mengapa?
(a) Internet of Things and DevOps
(b) Artificial Intelligence and DevOps

Security *in* DevOps Engineer

Why is security necessary for our cloud servers? Is there such a thing as a completely secure system?

Security is essential for the cloud servers due to several reasons. **<u>Cloud servers may store valuable and sensitive data</u>**, such as personal information and intellectual property. Without proper security measures, this data can be susceptible to unauthorized access, theft, or compromise.

Security *in* DevOps



Security *in* DevOps

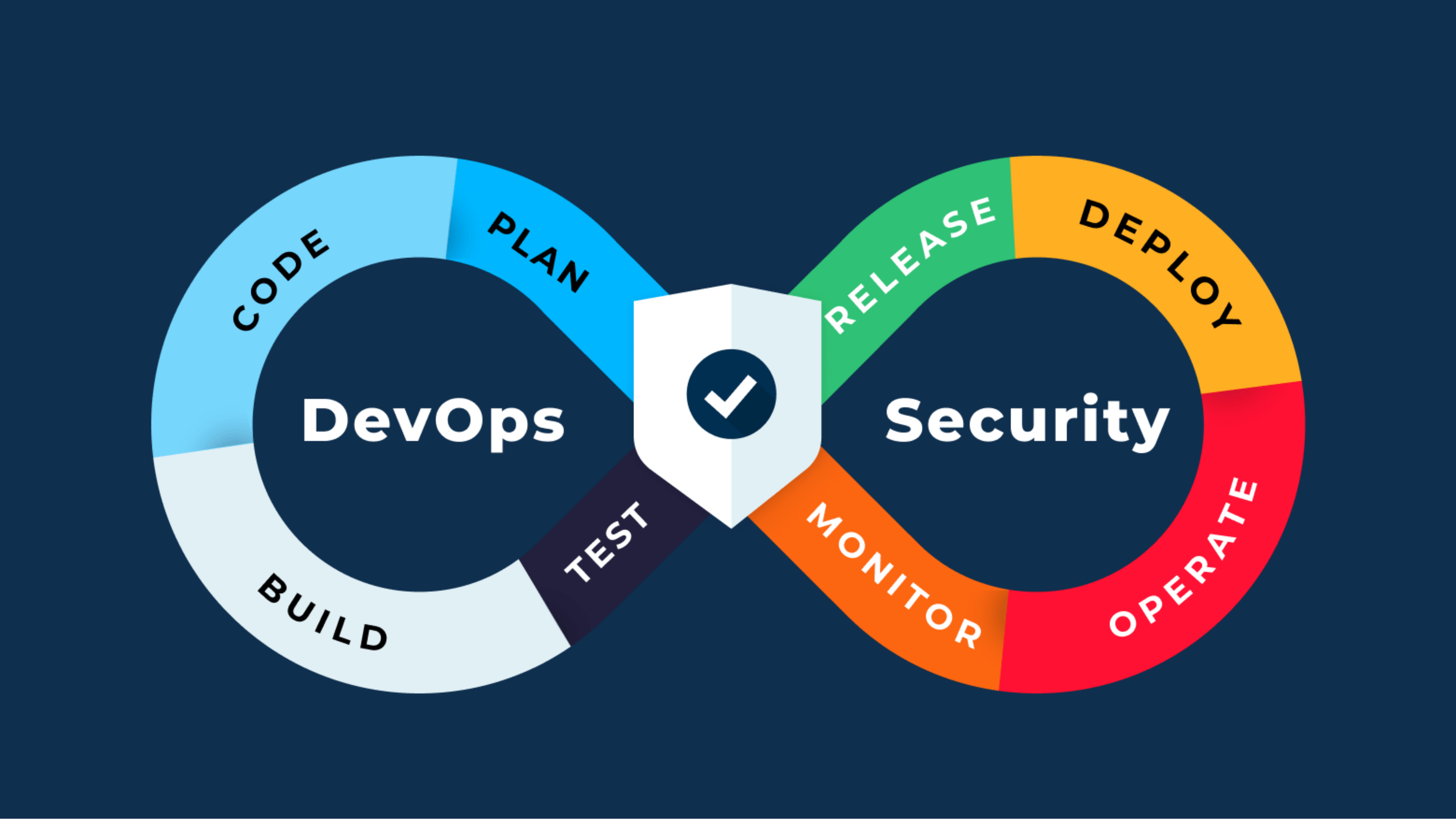
Pendekatan yang menggabungkan keamanan (*security*) dengan praktik pengembangan sistem/software yang cepat dan berkelanjutan dalam metodologi DevOps.

Tujuan utama dari Security in DevOps

To ensure that security is considered throughout the software development lifecycle, from design to product delivery. By implementing security from the start and automating security practices, organisations can reduce security risks, respond quickly to attacks, and ensure the reliability of the resulting systems.

DevOps security (*DevSecOps*) is an approach to cybersecurity that focuses on application development and development operations (*DevOps*). It combines three phrases:

1) Development
2) Operations
3) Security



DevOps teams are in no way exempt/free from the dangers of the **Digital Age**. With the number of *cyber threats growing all the time*, *increased alertness from clients* and customers, and *unsympathetic compliance regulations* to which companies are responsible, it is important for organizations to overlook the **potential dangers** within **DevOps pipelines**.

Why *is* DevOps Security Needed?

Important !!!

Security is one of the **critical areas of concern for developers**. Developers often rely heavily on Software Development Kits (SDKs) or at least on other frameworks or individual programs. But **third-party code** can represent a serious security vulnerability (*kerentanan keamanan*) in these cases. **As far as a developer is concerned, their organization doesn't necessarily know whether they've been addressed before.**

- **Libraries**.
- **Frameworks**.
- **Plugins dan Extensions**.
- **API Libraries**.

Third-party code adalah kode atau *software* yang dibuat oleh entitas atau pihak ketiga yang tidak terkait dengan tim pengembang atau organisasi. Third-party code biasanya tersedia sebagai komponen atau library untuk digunakan dalam pengembangan *software*.

Start *from* Here !!!

Start *with* New Cloud Server

1 – What *is* phpMyAdmin?

Next-level Database Management *in* IDCloudHost

(Connect PhpMyAdmin *to* Cloud SQL *in* Virtual Server)

[1] Install Apache2 *via* Putty

[1] Change hak akses *superuser* atau *root*:

- ***sudo su***

[2] Perbaharui paket yang tersedia ke versi terbaru ubuntu dengan perintah:

- ***sudo apt update***
- ***sudo apt upgrade***

[3] Install Apache di linux ubuntu 22.04, dengan menjalankan perintah berikut:

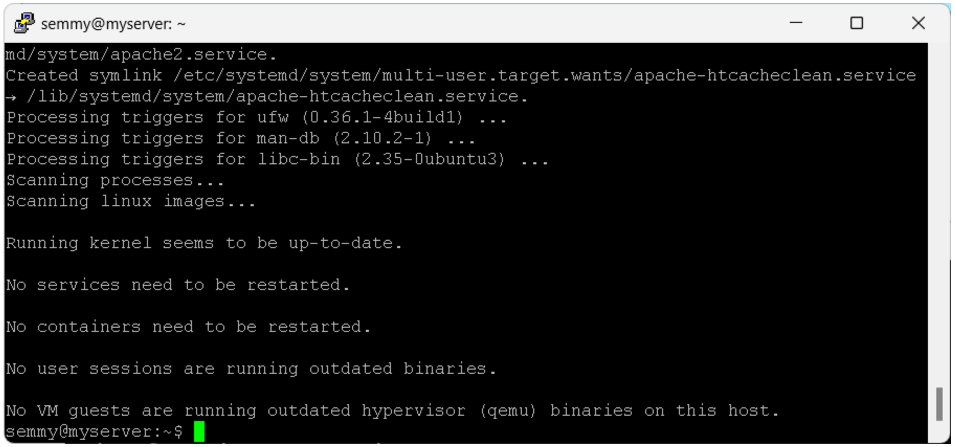
- ***sudo apt install apache2***

Keterangan:

1) Perintah ***sudo*** adalah singkatan dari "*Superuser Do*". Perintah ini digunakan untuk memberikan akses root atau superuser pada pengguna yang tidak memiliki izin tersebut.

2) Perintah ***su***, di sisi lain, adalah singkatan dari "*Substitute User*". Perintah ini digunakan untuk beralih ke akun pengguna lain tanpa harus keluar dari sesi terminal saat ini.

3) Perintah ***apt*** adalah singkatan dari "*Advanced Packaging Tool*". Perintah *apt* merupakan sebuah tool pada sistem operasi Debian dan turunannya, seperti Ubuntu, yang digunakan untuk mengelola paket software, yaitu menginstal, menghapus, atau memperbarui paket software pada sistem operasi.



[2] Install MySQL *in* Ubuntu Server

1. Untuk memulai instal MySQL bisa dimulai dengan perintah:

- ***sudo apt install mysql-server***

2. Selanjutnya aktifkan service MySQL di Ubuntu 22.04 dengan peritah:

- ***sudo systemctl start mysql.service***

3. Untuk menambahkan keamanan pada MySQL database, kita bisa menjalankan perintah ***mysql_secure_installation*** yang akan menambahkan beberapa pilihan seperti validasi ***keamanan password***, tidak memberikan akses untuk *remote login* dan membuang sample *user*.

Disini ada beberapa tahap yang harus kita lakukan, dimulai dari masuk ke MySQL prompt dengan mengetikan perintah berikut:

- ***sudo mysql***
- **mysql>** ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY '***ganti_password_kita***';

Kemudian keluar dari MySQL prompt

mysql> exit

[2] Install MySQL *in* Ubuntu Server

[4] Setelah metode authentikasi untuk *user root* kita ubah, kemudian kita jalankan perintah berikut:

- ***sudo mysql_secure_installation***

NOTE: *ADA BEBERAPA PERTANYAAN YANG AKAN DI JAWAB.*

[5] Setelah proses *mysql_secure_installation* selesai, selanjutnya kita bisa mengembalikan metoda authentikasi *user root* ke semula yaitu *auth_socket* dengan beberapa perintah seperti berikut:

> ***mysql -u root -p***

..

mysql> *ALTER USER 'root'@'localhost' IDENTIFIED WITH auth_socket;*

mysql> *FLUSH PRIVILEGES;*

mysql> *EXIT;*

[6] Selanjutnya, kita bisa melihat status MySQL service dengan perintah:

- ***systemctl status mysql.service***

[1] Install Latest Version *of* PHP

Opsi #1: menginstall PHP versi terbaru secara otomatis:

[1] Menginstall PHP sekaligus modul pendukung lainnya:

- *sudo apt install php libapache2-mod-php php-mbstring php-xmlrpc php-soap php-gd php-xml php-cli php-zip php-bcmath php-tokenizer php-json php-pear*

[2] Buat "*test.php*" file guna memastikan PHP sudah terinstall di server:

- *sudo nano /var/www/html/test.php*

Dan ketikan:

```
<?php
phpinfo();
?>
```

[3] Akhiri dengan klik tombol Ctrl + X dan type Y, lalu Enter secara berurutan.

[4] Silakan kembali ke browser dan ketikan "*http://72.168.19.80/test.php*". Browser akan memunculkan tampilan PHP version. Berarti kita sudah berhasil.

[2] Install Latest Version *of* PHP

Opsi #2: jika terjadi error pada cara pertama, gunakan cara yang kedua untuk menginstall PHP versi terbaru secara otomatis:

[1] Menginstall PHP sekaligus modul pendukung lainnya:
- *sudo apt install php libapache2-mod-php php-mysql*

[2] Buat "*test.php*" file guna memastikan PHP sudah terinstall di server:
- *sudo nano /var/www/html/test.php*

Dan ketikan:

```
<?php
phpinfo();
?>
```

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[3] Create Project Directory

[1] Masuk sebagai administrator atau *root*:

- *sudo su*

[2] Buat project folder baru dengan perintah (<u>Parent directory</u>):

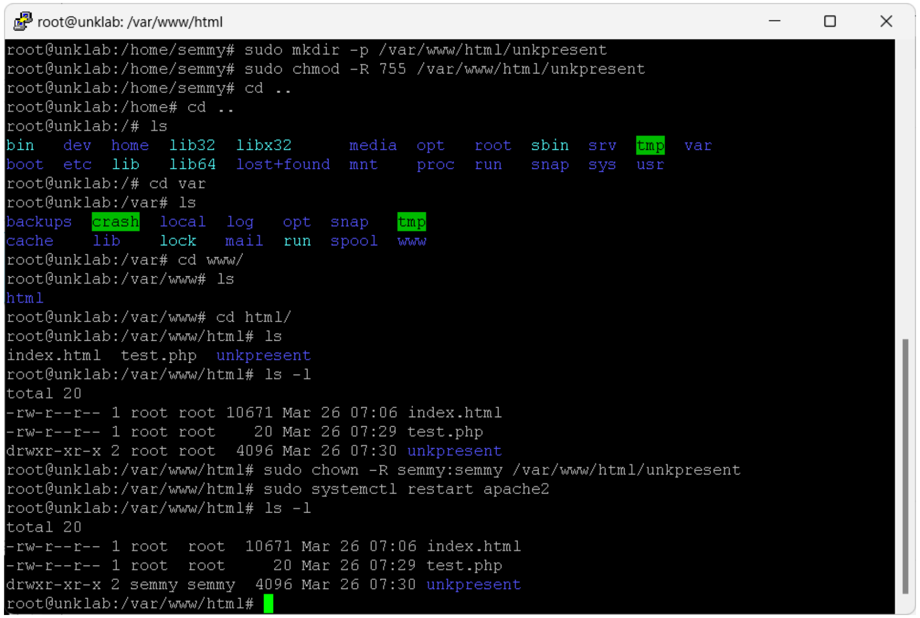
- *sudo mkdir -p /var/www/html/**unkpresent***

[3] Konfigurasi permisi hak akses dan pemilik dari folder:

- *sudo chmod -R 755 /var/www/html/**unkpresent***
- *sudo chown -R **<u>semmy:semmy</u>** /var/www/html/**unkpresent***

[4] Restart Apache Web Server:

- *sudo systemctl restart apache2*



[4] Config File .htaccess *in* Apache2

Untuk mengaktifkan file *.htaccess* di Apache2 pada Ubuntu Server, ikuti langkah-langkah berikut:

[1] Pastikan modul rewrite sudah diaktifkan. Anda dapat memeriksa apakah modul ini sudah diaktifkan dengan menjalankan perintah:

- ***sudo a2enmod rewrite***

[2] Setelah modul rewrite diaktifkan, buka file konfigurasi untuk situs web yang ingin Anda aktifkan .htaccess. Misalnya, jika Anda ingin mengaktifkan .htaccess untuk situs web default, buka file konfigurasinya dengan menjalankan perintah:

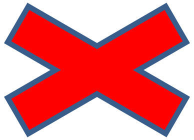
- ***sudo nano /etc/apache2/sites-available/000-default.conf***

[3] Di dalam blok konfigurasi untuk situs web, tambahkan konfigurasi berikut di bawah direktif **<VirtualHost>**:

<Directory "/var/www/html">
AllowOverride All
</Directory>

[4] **OR** Jika Anda ingin mengaktifkan .htaccess di direktori /home/user/public_html, maka pada langkah ke-3, gantilah konfigurasi sebagai berikut:

<Directory "/home/semmy/public_html">
AllowOverride All
</Directory>



[5] Simpan dan tutup file konfigurasi dengan menekan **Ctrl+X**, lalu ketik **Y** untuk menyimpan perubahan dan Enter untuk menutup editor.

[6] Restart apache web server:

- ***sudo systemctl restart apache2***

What *is* phpMyAdmin?

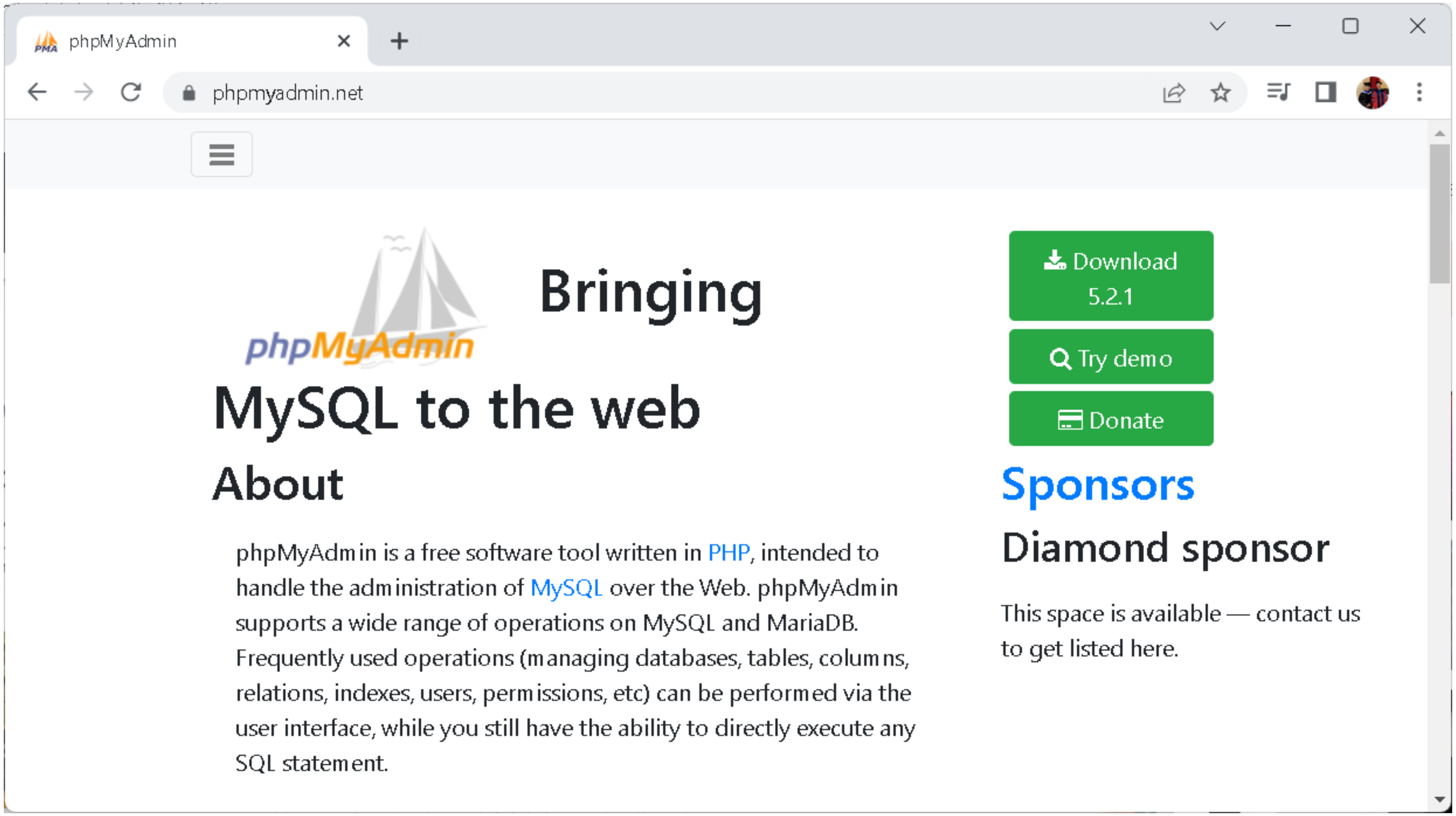
- **phpMyAdmin** is a web-based tool used for **managing MySQL databases**. It provides an **easy-to-use graphical interface for performing various database operations**, such as creating, deleting, and modifying databases, tables, and fields.

- On a *virtual server* provided by idCloudHost, we can install and use phpMyAdmin to manage your MySQL databases. To do so, we need to first install a web server like Apache or Nginx and then install and configure phpMyAdmin on top of it.

- Once installed, we can access phpMyAdmin by navigating to its URL in our web browser. The exact URL will depend on our server configuration, but it will typically be something like ***http://your-server-ip/phpmyadmin***.

- Once we have accessed phpMyAdmin, we can log in using our MySQL database credentials and begin managing our databases.

PhpMyAdmin *for* MySQL



Option #1

install phpMyAdmin

How To Install **phpMyAdmin** *on* Ubuntu 22.04???

Install phpMyAdmin *on* Ubuntu 22.04

Download the latest phpMyAdmin archive from the official download page:

- sudo **su**
- sudo apt install **unzip**
- wget https://files.phpmyadmin.net/phpMyAdmin/5.2.0/phpMyAdmin-5.2.0-all-languages.zip
- unzip phpMyAdmin-5.2.0-all-languages.zip
- sudo **mv** phpMyAdmin-5.2.0-all-languages /usr/share/phpmyadmin

Create “tmp directory” and set the proper permissions:

- sudo mkdir /usr/share/phpmyadmin/tmp
- sudo chown -R www-data:www-data /usr/share/phpmyadmin
- sudo chmod 777 /usr/share/phpmyadmin/tmp

Unzip package belum di install:

```
root@julioserver:/home/jmsgi# unzip phpMyAdmin-5.2.1-all-languages.zip
Command 'unzip' not found, but can be installed with:
apt install unzip
root@julioserver:/home/jmsgi# sudo mv phpMyAdmin-5.2.1-all-languages.zip /usr/sh
are/phpmyadmin
mv: cannot stat 'phpMyAdmin-5.2.1-all-languages.zip': No such file or directory
```

Configure phpMyAdmin

Create an Apache configuration file for phpMyAdmin:

- *sudo vim /etc/apache2/conf-available/phpmyadmin.conf*

Add the below content to the file:

```
Alias /phpMyAdmin /usr/share/phpmyadmin

<Directory /usr/share/phpmyadmin/>
AddDefaultCharset UTF-8
<IfModule mod_authz_core.c>
<RequireAny>
Require all granted
</RequireAny>
</IfModule>
</Directory>

<Directory /usr/share/phpmyadmin/setup/>
<IfModule mod_authz_core.c>
<RequireAny>
Require all granted
</RequireAny>
</IfModule>
</Directory>
```

Simpan file tersebut dengan cara tekan tombol **ESC** untuk beralih ke perintah lainnya. Kemudian ketik **:wq** (titik dua+w+q) dan tekan tombol Enter.

Restart the Apache service to reload all settings:

- *sudo a2enconf phpmyadmin*
- *sudo systemctl restart apache2*
- *sudo systemctl reload apache2*
- *sudo systemctl status apache2.service*

Create MySQL Database & User

Connect to the MySQL server running on your system:

- *sudo mysql*
- *mysql -u root -p*

Create a *database* and *user*, and assign the privileges to the user on the database:

Note*: Ubah user "iscdbadmin" dengan nama user kalian sendiri.

mysql> CREATE DATABASE iscdbadmin;
Query OK, 1 row affected (0.01 sec)

mysql> CREATE USER 'iscdbadmin'@'localhost' IDENTIFIED BY 'NewPa$$w0rd';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT ALL ON iscdbadmin.* TO 'iscdbadmin'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT ALL PRIVILEGES ON *.* TO '*iscdbadmin*'@'localhost';
Query OK, 0 rows affected (0.71 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)

Pada saat masuk ke phpmyadmin via browser, ada kemungkinan belum bisa untuk create new database.

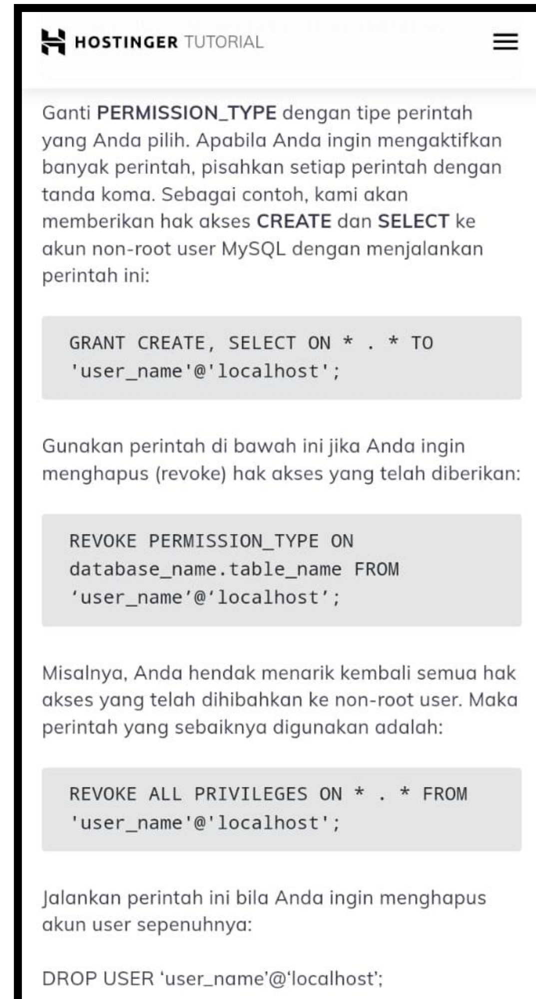
Grant User Permissions

Perintah SQL yang digunakan untuk memberikan izin kepada user 'testadmin' yang terhubung dari host 'localhost' untuk membuat objek (tabel, basis data, dll.) di semua basis data ('.' dalam konteks ini berarti semua basis data).

$ sudo mysql -u root -p

$ GRANT ALL PRIVILEGES ON *.* TO 'root'@'localhost';
$ GRANT ALL PRIVILEGES ON *.* TO '**testadmin**'@'localhost';
~~$ GRANT create on *.* to '**testadmin**@localhost';~~
~~$ GRANT create on *.* to '**root**@localhost';~~
~~$ GRANT ALL PRIVILEGES ON *.* TO '**root**'@'localhost';~~

$ FLUSH PRIVILEGES;



HOSTINGER TUTORIAL

Ganti **PERMISSION_TYPE** dengan tipe perintah yang Anda pilih. Apabila Anda ingin mengaktifkan banyak perintah, pisahkan setiap perintah dengan tanda koma. Sebagai contoh, kami akan memberikan hak akses **CREATE** dan **SELECT** ke akun non-root user MySQL dengan menjalankan perintah ini:

```
GRANT CREATE, SELECT ON * . * TO
'user_name'@'localhost';
```

Gunakan perintah di bawah ini jika Anda ingin menghapus (revoke) hak akses yang telah diberikan:

```
REVOKE PERMISSION_TYPE ON
database_name.table_name FROM
'user_name'@'localhost';
```

Misalnya, Anda hendak menarik kembali semua hak akses yang telah dihibahkan ke non-root user. Maka perintah yang sebaiknya digunakan adalah:

```
REVOKE ALL PRIVILEGES ON * . * FROM
'user_name'@'localhost';
```

Jalankan perintah ini bila Anda ingin menghapus akun user sepenuhnya:

DROP USER 'user_name'@'localhost';

Don't forget to:

1. Clear your browser cookies/history
2. Logout and login again to the phpMyAdmin

Change User Root Password

Perintah SQL yang digunakan untuk mengatur password untuk user "*root*" MySQL sebagai tujuan dalam meningkatkan keamanan sistem database di PHPmyAdmin:

[1] Login ke MySQL sebagai Pengguna root Tanpa Password:

- *sudo mysql -u root*

[2] Set Password untuk Pengguna root (Ganti *'newpassword*' dengan kata sandi):

- *ALTER USER 'root'@'localhost' IDENTIFIED WITH 'mysql_native_password' BY 'newpassword';*

[3] Perbarui Izin dan Reload Hak Akses:

- *FLUSH PRIVILEGES;*

[4] Keluar dari Shell MySQL:

- *Exit*

[5] Uji Login dengan Kata Sandi Baru:

- *mysql -u root –p*

More Settings

[1] Periksa Alias dalam Konfigurasi Apache:
- sudo su
- nano /etc/apache2/conf-available/phpmyadmin.conf

Pastikan bahwa alias untuk phpMyAdmin sudah benar di file /etc/apache2/conf-available/phpmyadmin.conf.

Isi file tersebut seharusnya mirip dengan di samping ini:

```
Alias /phpmyadmin /usr/share/phpmyadmin

<Directory /usr/share/phpmyadmin>
  Options FollowSymLinks
  DirectoryIndex index.php

  <IfModule mod_php5.c>
    <IfModule mod_mime.c>
      AddType application/x-httpd-php .php
    </IfModule>
    <FilesMatch ".+\.php$">
      SetHandler application/x-httpd-php
    </FilesMatch>
  </IfModule>

  <IfModule mod_php7.c>
    <IfModule mod_mime.c>
      AddType application/x-httpd-php .php
    </IfModule>
    <FilesMatch ".+\.php$">
      SetHandler application/x-httpd-php
    </FilesMatch>
  </IfModule>

  <IfModule mod_php.c>
    <IfModule mod_mime.c>
      AddType application/x-httpd-php .php
    </IfModule>
    <FilesMatch ".+\.php$">
      SetHandler application/x-httpd-php
    </FilesMatch>
  </IfModule>

  <IfModule mod_deflate.c>
    <IfModule mod_filter.c>
      AddOutputFilterByType DEFLATE application/json
      AddOutputFilterByType DEFLATE application/vnd.ms-excel
      AddOutputFilterByType DEFLATE application/vnd.ms-powerpoint
      AddOutputFilterByType DEFLATE application/vnd.ms-word
      AddOutputFilterByType DEFLATE application/vnd.oasis.opendocument.presentation
      AddOutputFilterByType DEFLATE application/vnd.oasis.opendocument.spreadsheet
      AddOutputFilterByType DEFLATE application/vnd.oasis.opendocument.text
      AddOutputFilterByType DEFLATE application/vnd.openxmlformats-officedocument.presentationml.presentation
      AddOutputFilterByType DEFLATE application/vnd.openxmlformats-officedocument.spreadsheetml.sheet
      AddOutputFilterByType DEFLATE application/vnd.openxmlformats-officedocument.wordprocessingml.document
      AddOutputFilterByType DEFLATE application/x-javascript
      AddOutputFilterByType DEFLATE application/xhtml+xml
      AddOutputFilterByType DEFLATE application/xml
      AddOutputFilterByType DEFLATE image/svg+xml
      AddOutputFilterByType DEFLATE text/css
      AddOutputFilterByType DEFLATE text/html
      AddOutputFilterByType DEFLATE text/javascript
      AddOutputFilterByType DEFLATE text/plain
    </IfModule>
  </IfModule>

  <IfModule mod_authz_core.c>
    <RequireAny>
      Require all granted
    </RequireAny>
  </IfModule>
  <IfModule !mod_authz_core.c>
    Order Allow,Deny
    Allow from all
    Satisfy Any
  </IfModule>

</Directory>
```

[2] Periksa Modul-Modul Apache:
- sudo a2enmod php
- sudo a2enmod rewrite
- sudo systemctl restart apache2

[3] Periksa Permissions:
- sudo chown -R www-data:www-data /usr/share/phpMyAdmin
- sudo chmod -R 755 /usr/share/phpmyadmin

Access phpMyAdmin

Open web browser to access phpMyAdmin with the server IP address or domain name:
> *http://your-server-ip-domain/phpmyadmin*

```
http://your-server-ip-domain/phpmyadmin
```

Option #2

install phpMyAdmin

How *to* install **phpMyAdmin** as a web-based tool used for managing MySQL databases on Ubuntu 22.04?

How *to* Install PhpMyAdmin?

Install di ubuntu 22.04 dengan perintah (*akan ada beberapa pertanyaan*):

- *sudo apt install phpmyadmin*

Duplikasi konfigurasi PhpMyAdmin untuk Apache web server:

- sudo cp /etc/phpmyadmin/apache.conf /etc/apache2/conf-available/phpmyadmin.conf

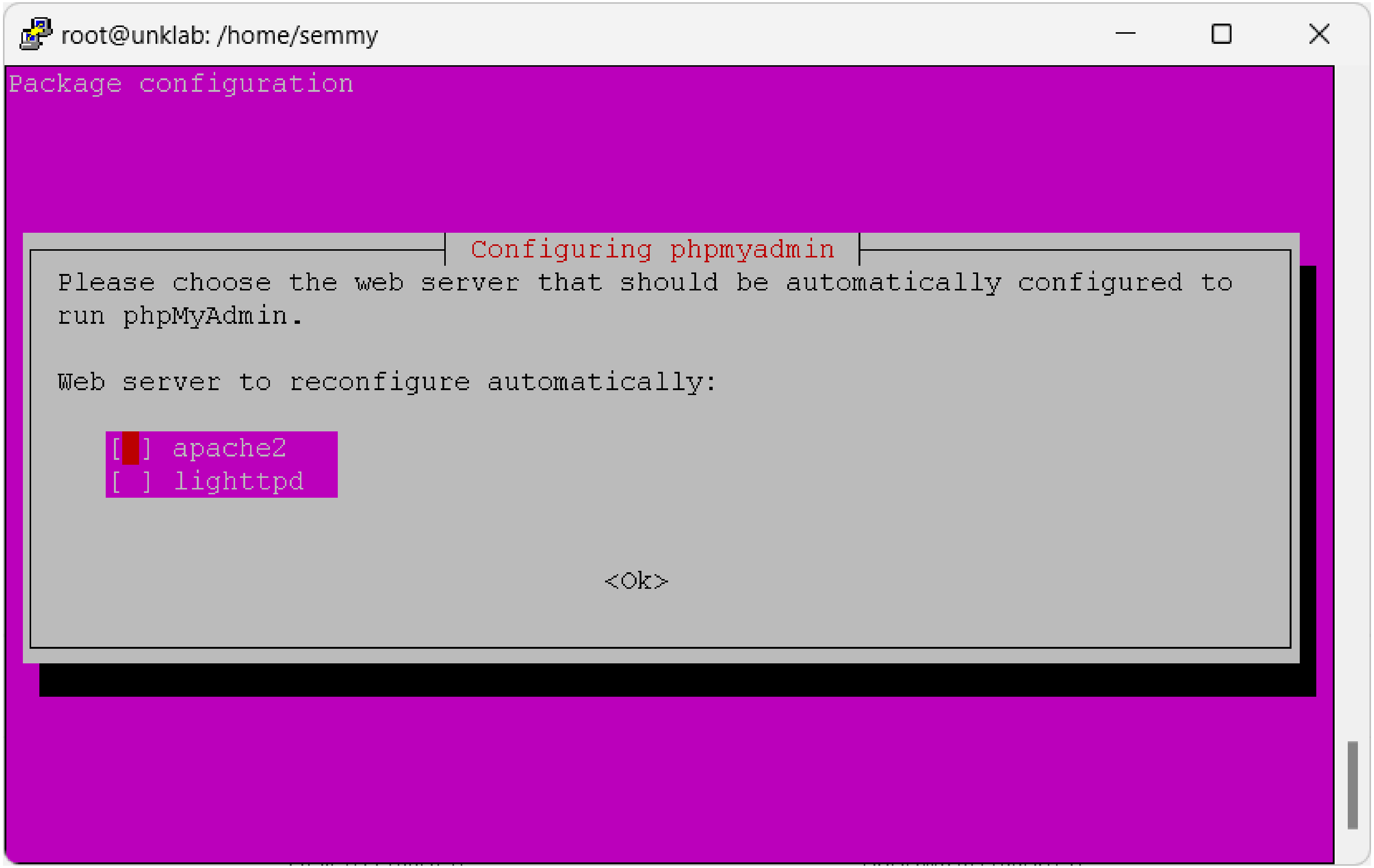
Selanjutnya **enable konfigurasi** tersebut dengan perintah:

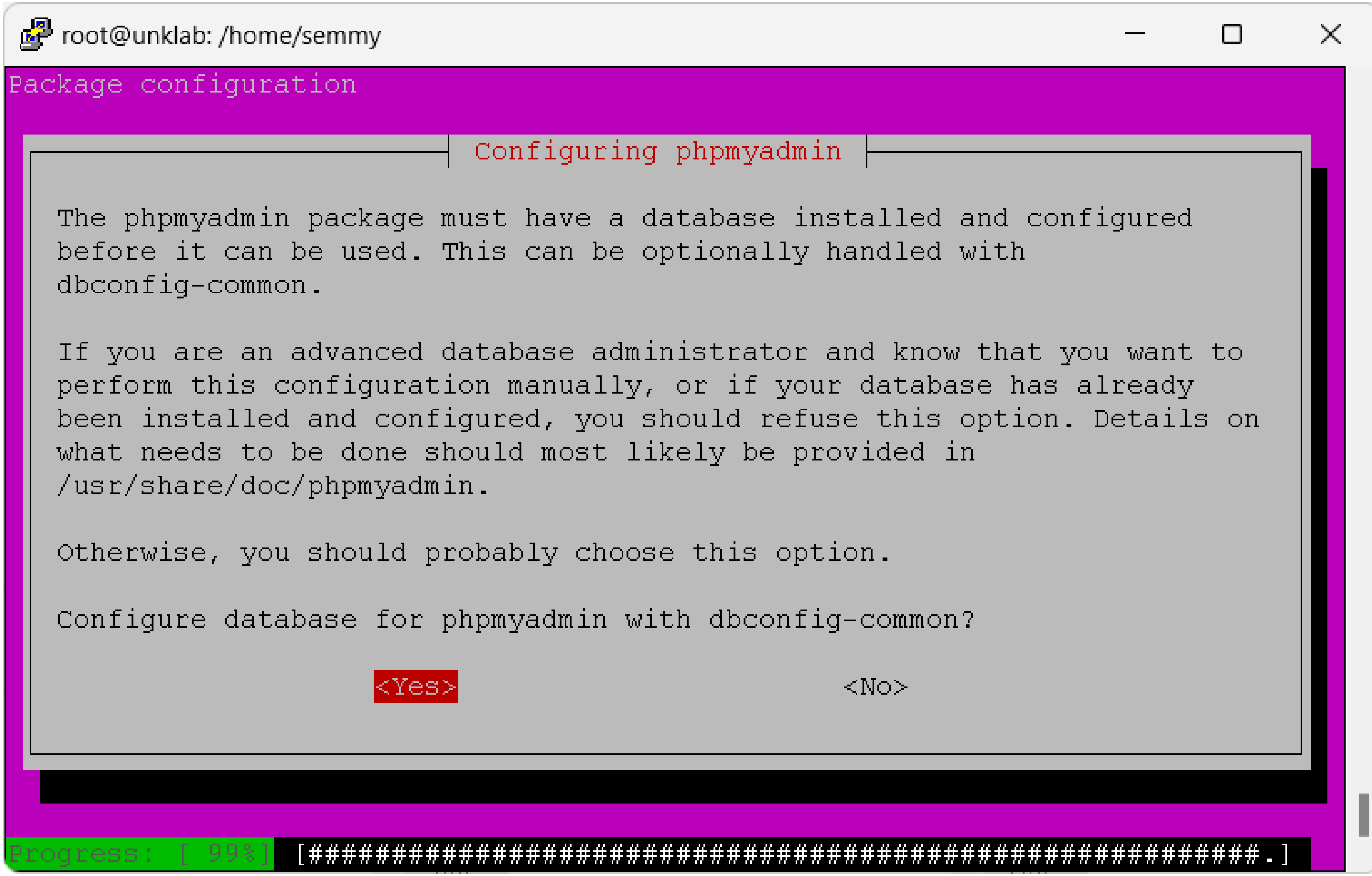
- *sudo a2enconf phpmyadmin.conf*

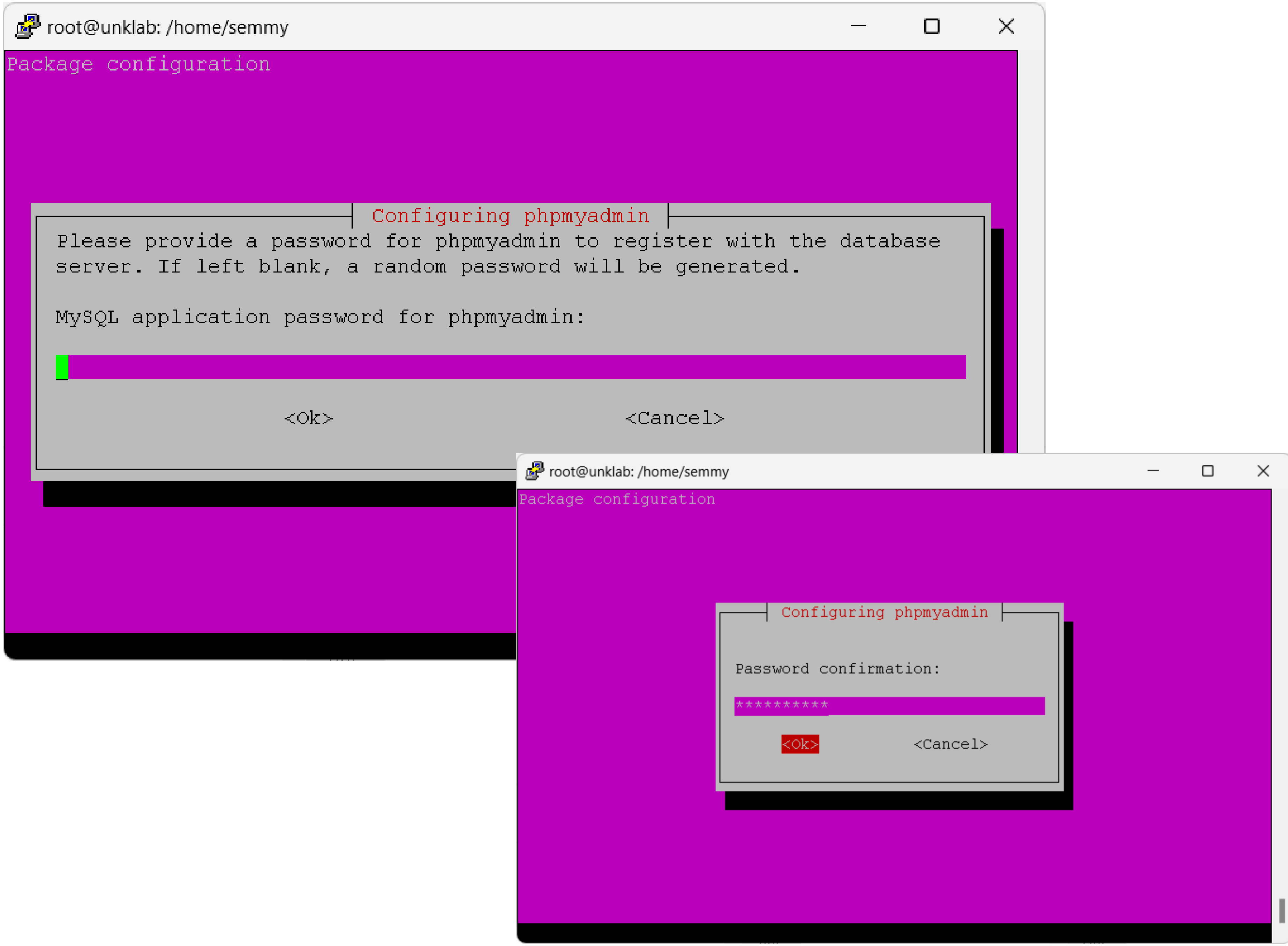
Jangan lupa restart Apache:

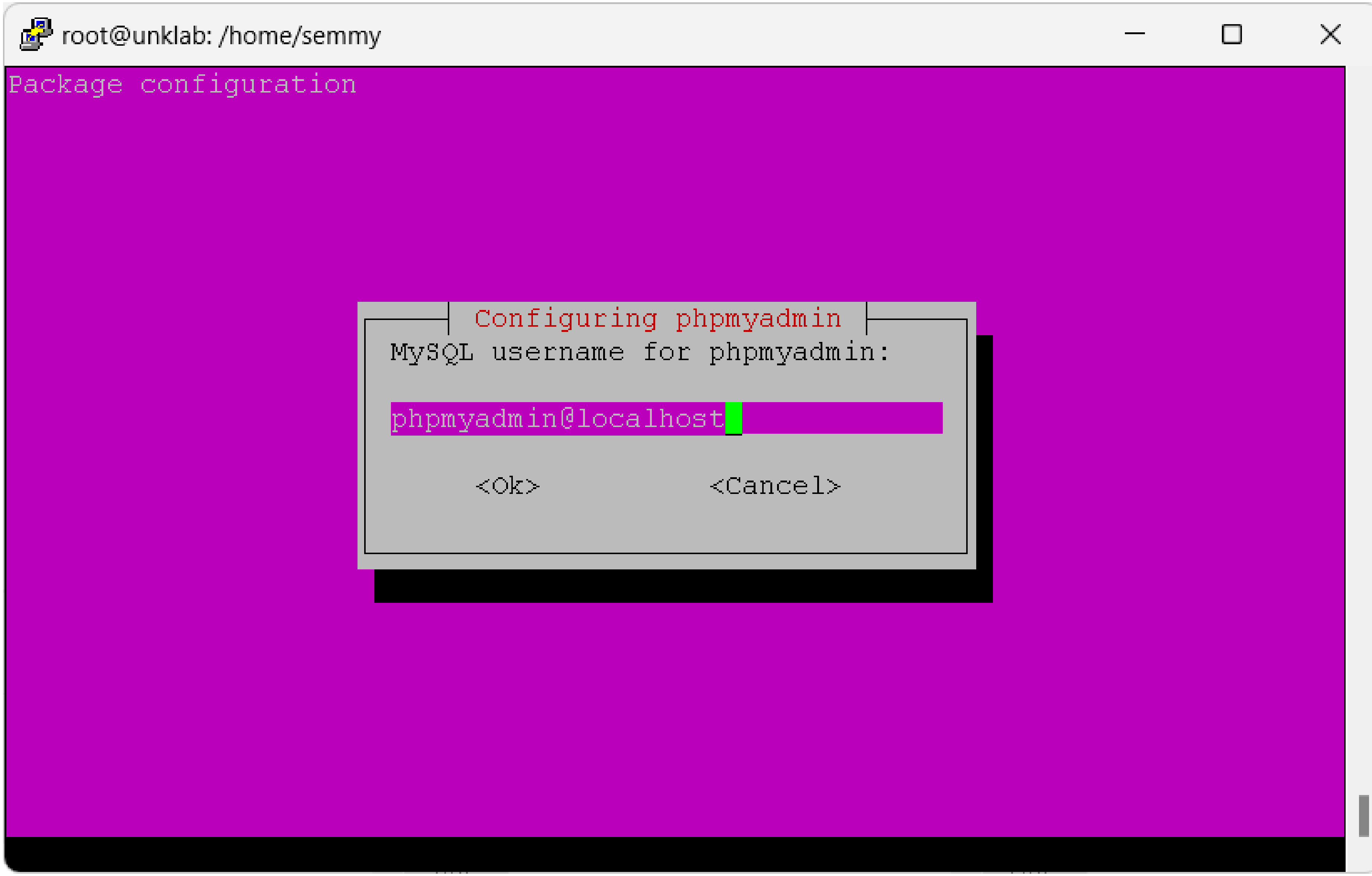
- *sudo service apache2 restart*
- *sudo systemctl reload apache2*

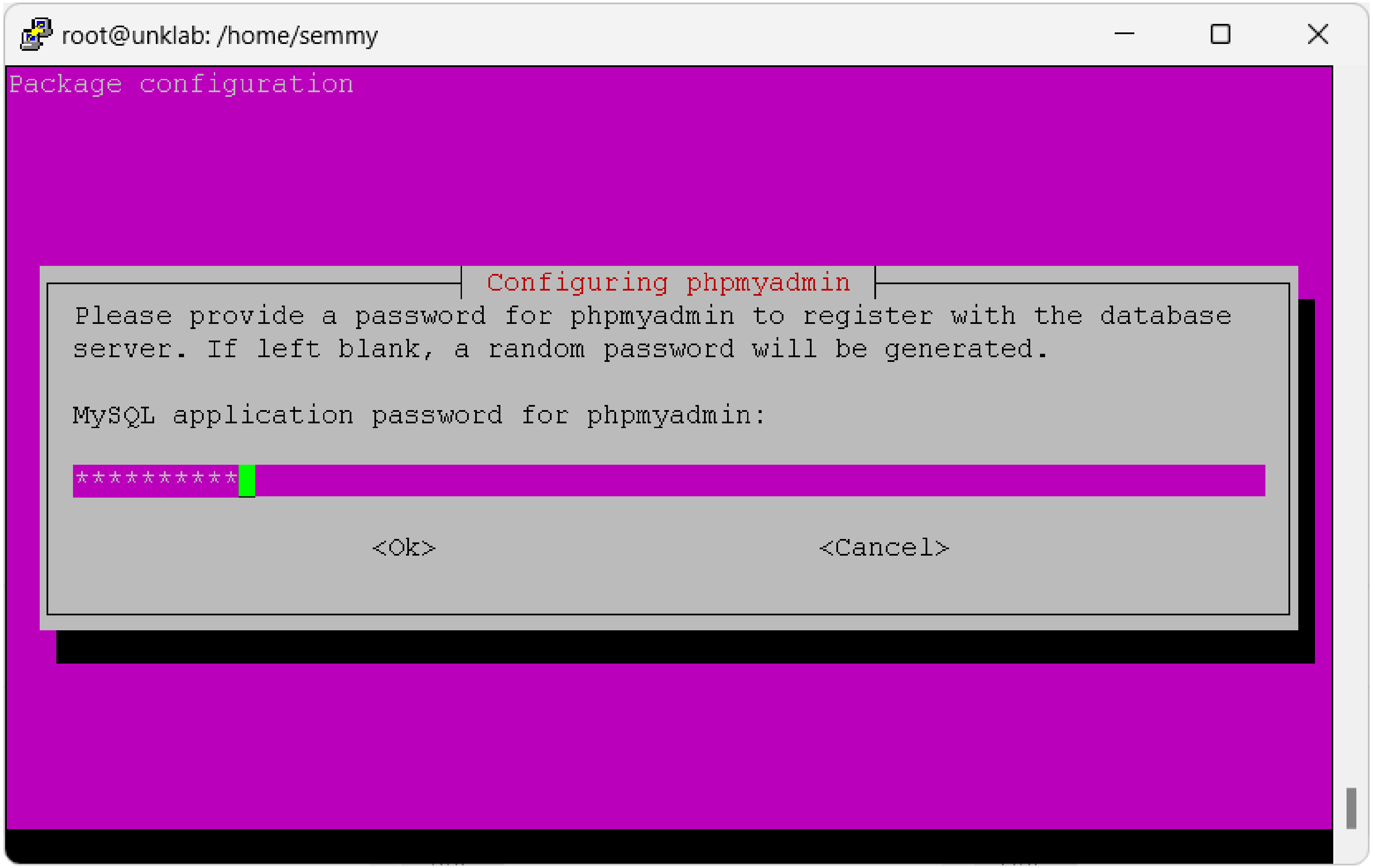
Kita sudah bisa melakukan akses dengan memasukan alamat domain (*public IP address*) dan ditambah /phpmyadmin.
Contoh access URL: *http://103.49.239.80/phpmyadmin/*


root@unklab: /home/semmy
Package configuration
Configuring phpmyadmin
Please choose the web server that should be automatically configured to run phpMyAdmin.
Web server to reconfigure automatically:
[ ] apache2
[ ] lighttpd
<Ok>


root@unklab: /home/semmy
Package configuration
Configuring phpmyadmin
The phpmyadmin package must have a database installed and configured before it can be used. This can be optionally handled with dbconfig-common.
If you are an advanced database administrator and know that you want to perform this configuration manually, or if your database has already been installed and configured, you should refuse this option. Details on what needs to be done should most likely be provided in /usr/share/doc/phpmyadmin.
Otherwise, you should probably choose this option.
Configure database for phpmyadmin with dbconfig-common?
<Yes>
<No>
Progress: [99%]
[###.]


root@unklab: /home/semmy
Package configuration
Configuring phpmyadmin
Please provide a password for phpmyadmin to register with the database server. If left blank, a random password will be generated.
MySQL application password for phpmyadmin:
<Ok>
<Cancel>
root@unklab: /home/semmy
Package configuration
Configuring phpmyadmin
Password confirmation:
****
<Ok>
<Cancel>

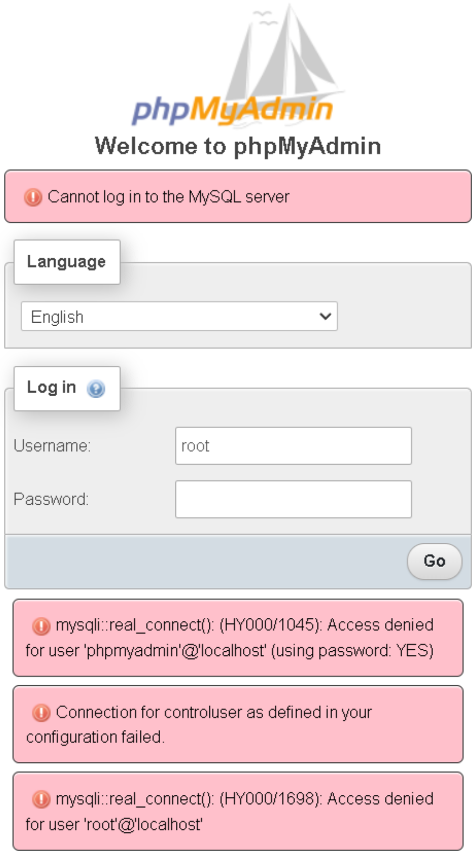

root@unklab: /home/semmy
Package configuration
Configuring phpmyadmin
MySQL username for phpmyadmin:
phpmyadmin@localhost
<Ok>
<Cancel>



Jika muncul error pada saat membuat password dari user PhpMyAdmin. Nanti bisa di *ignore* saja.

Access Denied *for* User “phpmyadmin”

Error: mysqli::real_connect(): (HY000/1045): Access denied for user 'phpmyadmin'@'localhost' (using password: YES)



Change to super user:

- *sudo su*

Access MySQL:

- *mysql -u root -p*

Create User phpmyadmin with Password:

- CREATE USER 'phpmyadmin'@'localhost' IDENTIFIED BY '**1234567890**';
- GRANT ALL PRIVILEGES ON *.* TO 'phpmyadmin'@'localhost' WITH GRANT OPTION;
- FLUSH PRIVILEGES;
- exit

More Settings

[1] Periksa Alias dalam Konfigurasi Apache:

- sudo su
- nano /etc/apache2/conf-available/phpmyadmin.conf

Pastikan bahwa alias untuk phpMyAdmin sudah benar di file /etc/apache2/conf-available/phpmyadmin.conf.

Isi file tersebut seharusnya mirip dengan di samping ini:

```
Alias /phpmyadmin /usr/share/phpmyadmin

<Directory /usr/share/phpmyadmin>
  Options FollowSymLinks
  DirectoryIndex index.php

  <IfModule mod_php5.c>
    <IfModule mod_mime.c>
      AddType application/x-httpd-php .php
    </IfModule>
    <FilesMatch ".+\.php$">
      SetHandler application/x-httpd-php
    </FilesMatch>
  </IfModule>

  <IfModule mod_php7.c>
    <IfModule mod_mime.c>
      AddType application/x-httpd-php .php
    </IfModule>
    <FilesMatch ".+\.php$">
      SetHandler application/x-httpd-php
    </FilesMatch>
  </IfModule>

  <IfModule mod_php.c>
    <IfModule mod_mime.c>
      AddType application/x-httpd-php .php
    </IfModule>
    <FilesMatch ".+\.php$">
      SetHandler application/x-httpd-php
    </FilesMatch>
  </IfModule>

  <IfModule mod_deflate.c>
    <IfModule mod_filter.c>
      AddOutputFilterByType DEFLATE application/json
      AddOutputFilterByType DEFLATE application/vnd.ms-excel
      AddOutputFilterByType DEFLATE application/vnd.ms-powerpoint
      AddOutputFilterByType DEFLATE application/vnd.ms-word
      AddOutputFilterByType DEFLATE application/vnd.oasis.opendocument.presentation
      AddOutputFilterByType DEFLATE application/vnd.oasis.opendocument.spreadsheet
      AddOutputFilterByType DEFLATE application/vnd.oasis.opendocument.text
      AddOutputFilterByType DEFLATE application/vnd.openxmlformats-officedocument.presentationml.presentation
      AddOutputFilterByType DEFLATE application/vnd.openxmlformats-officedocument.spreadsheetml.sheet
      AddOutputFilterByType DEFLATE application/vnd.openxmlformats-officedocument.wordprocessingml.document
      AddOutputFilterByType DEFLATE application/x-javascript
      AddOutputFilterByType DEFLATE application/xhtml+xml
      AddOutputFilterByType DEFLATE application/xml
      AddOutputFilterByType DEFLATE image/svg+xml
      AddOutputFilterByType DEFLATE text/css
      AddOutputFilterByType DEFLATE text/html
      AddOutputFilterByType DEFLATE text/javascript
      AddOutputFilterByType DEFLATE text/plain
    </IfModule>
  </IfModule>

  <IfModule mod_authz_core.c>
    <RequireAny>
      Require all granted
    </RequireAny>
  </IfModule>
  <IfModule !mod_authz_core.c>
    Order Allow,Deny
    Allow from all
    Satisfy Any
  </IfModule>

</Directory>
```

[2] Periksa Modul-Modul Apache:

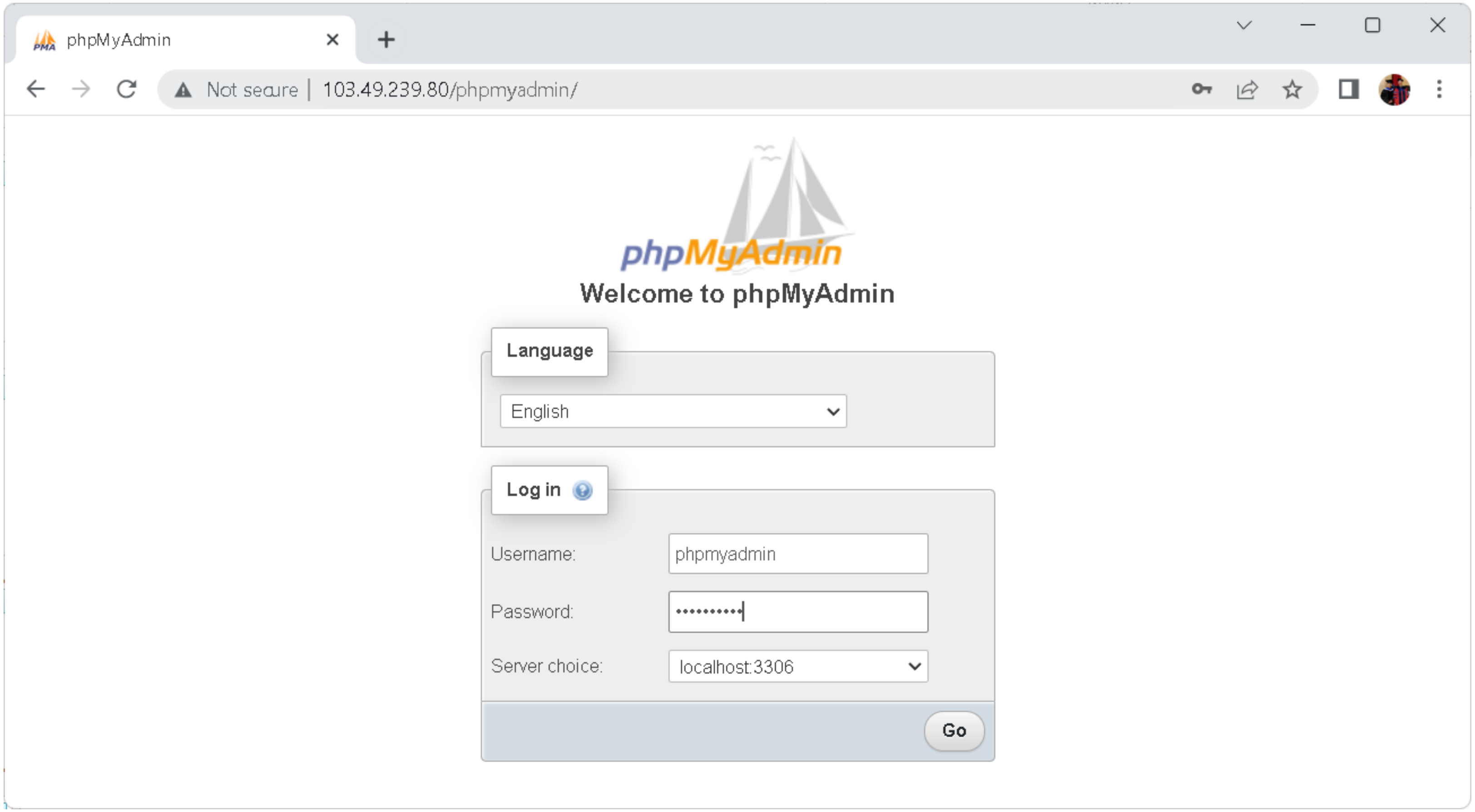
- sudo a2enmod php
- sudo a2enmod rewrite
- sudo systemctl restart apache2

[3] Periksa Permissions:

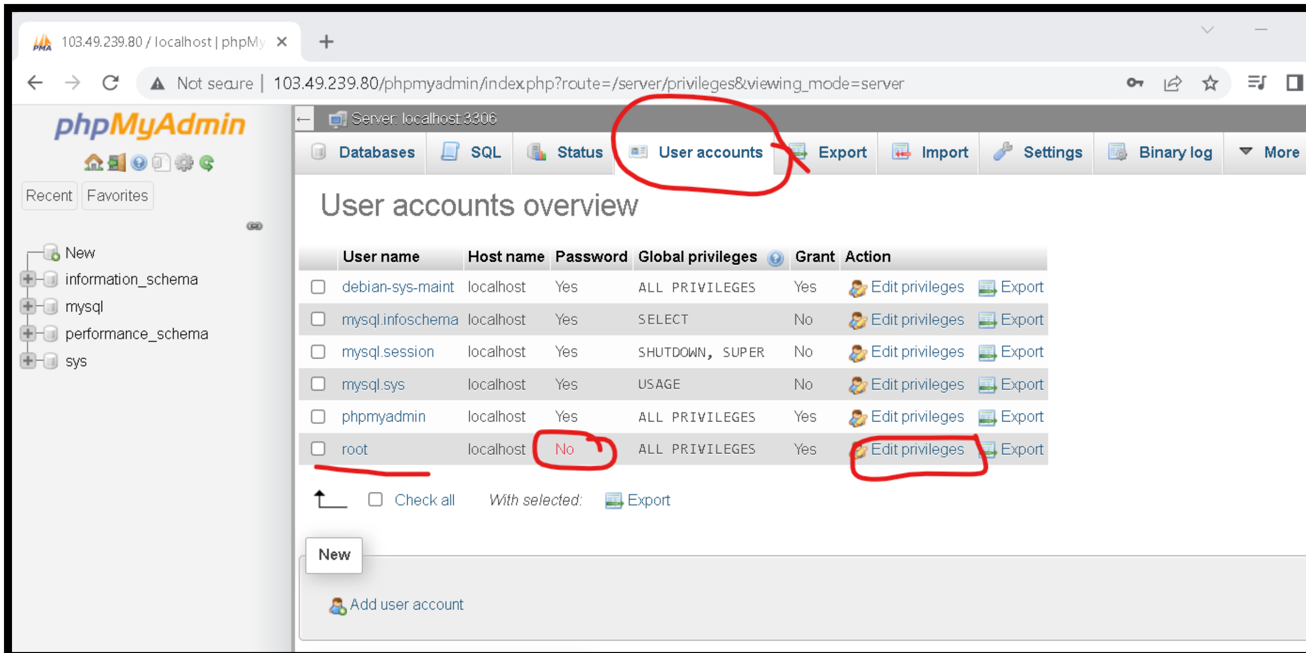
- sudo chown -R www-data:www-data /usr/share/phpMyAdmin
- sudo chmod -R 755 /usr/share/phpmyadmin

Login PhpMyAdmin

Access URL : *http://103.49.239.80/phpmyadmin/*



User Root *vs* User Phpmyadmin



Perbedaan antara ***user root*** dan ***user phpmyadmin*** terletak pada hak akses yang dimilikinya. <u>*Root user*</u> memiliki akses penuh ke sistem dan database MySQL, sementara ***user phpmyadmin*** hanya memiliki akses ke MySQL melalui aplikasi web phpMyAdmin.

Problem When Install “PhpMyAdmin”

To (Un)Install phpMyAdmin on Ubuntu 22.04

sudo apt-get remove phpmyadmin

Mengonfigurasi ulang **paket phpMyAdmin** pada sistem operasi berbasis Debian atau Ubuntu:

sudo dpkg-reconfigure phpmyadmin

Problem When Install “PhpMyAdmin”

Ada beberapa kemungkinan mengapa PHPmyAdmin tidak dapat dibuka melalui browser setelah proses instalasi yang tepat:

1) **<u>Kesalahan dalam konfigurasi Apache:</u>** Pastikan bahwa konfigurasi PhpMyAdmin telah dikonfigurasi dengan benar pada Apache web server, dengan perintah "**sudo cp /etc/phpmyadmin/apache.conf /etc/apache2/conf-available/phpmyadmin.conf**" dan "**sudo a2enconf phpmyadmin.conf**". Selain itu, pastikan bahwa Apache web server telah di-restart setelah mengaktifkan konfigurasi PhpMyAdmin dengan perintah "**sudo service apache2 restart**".

2) **<u>Masalah dalam file konfigurasi PhpMyAdmin:</u>** Periksa apakah file konfigurasi PhpMyAdmin pada "**/etc/phpmyadmin/config.inc.php**" sudah dikonfigurasi dengan benar, terutama pada pengaturan koneksi ke server database.

3) **<u>Masalah dalam instalasi PhpMyAdmin:</u>** Coba lakukan proses instalasi ulang PhpMyAdmin dengan perintah "**sudo apt-get remove phpmyadmin**" diikuti dengan "**sudo apt-get install phpmyadmin**".

4) **<u>Firewall yang menghalangi akses:</u>** Pastikan bahwa firewall pada sistem Anda tidak memblokir akses ke PhpMyAdmin.

5) **<u>Masalah browser atau cache:</u>** Cobalah untuk membuka PhpMyAdmin pada browser yang berbeda atau membersihkan cache pada browser yang sedang digunakan.

6) **<u>Masalah dalam server database:</u>** Pastikan bahwa server database yang digunakan oleh PhpMyAdmin telah diaktifkan dan berjalan dengan baik.

Check Apache Error

Perintah ini (*sudo nano /var/log/apache2/error.log*) digunakan untuk membuka berkas log kesalahan Apache pada sistem Linux. Berkas log kesalahan (*error.log*) adalah tempat di mana Apache menyimpan catatan tentang kesalahan yang terjadi saat menjalankan layanan web.

> sudo nano /var/log/apache2/error.log

(IMPORTANT)

How To Install Linux, Apache, MySQL, PHP (LAMP) stack on Ubuntu 22.04

Link: https://www.digitalocean.com/community/tutorials/how-to-install-linux-apache-mysql-php-lamp-stack-on-ubuntu-22-04

Firewall settings to allow HTTP traffic:

```
$ sudo ufw app list
```

Copy

```
Output
Available applications:
  Apache
  Apache Full
  Apache Secure
  OpenSSH
```

- Apache: This profile opens only port 80 (normal, unencrypted web traffic).
- Apache Full: This profile opens both port 80 (normal, unencrypted web traffic) and port 443 (Transport Layer Security/Secure Sockets Layer (TLS/SSL) encrypted traffic).
- Apache Secure: This profile opens only port 443 (TLS/SSL encrypted traffic).
- OpenSSH pada Ubuntu Server digunakan untuk mengamankan akses jarak jauh ke sistem melalui protokol SSH (Secure Shell).

To allow traffic on port 80, use Apache profile:

```
$ sudo ufw allow in "Apache"
```

Copy

Verify the change with:

```
$ sudo ufw status
```

Copy

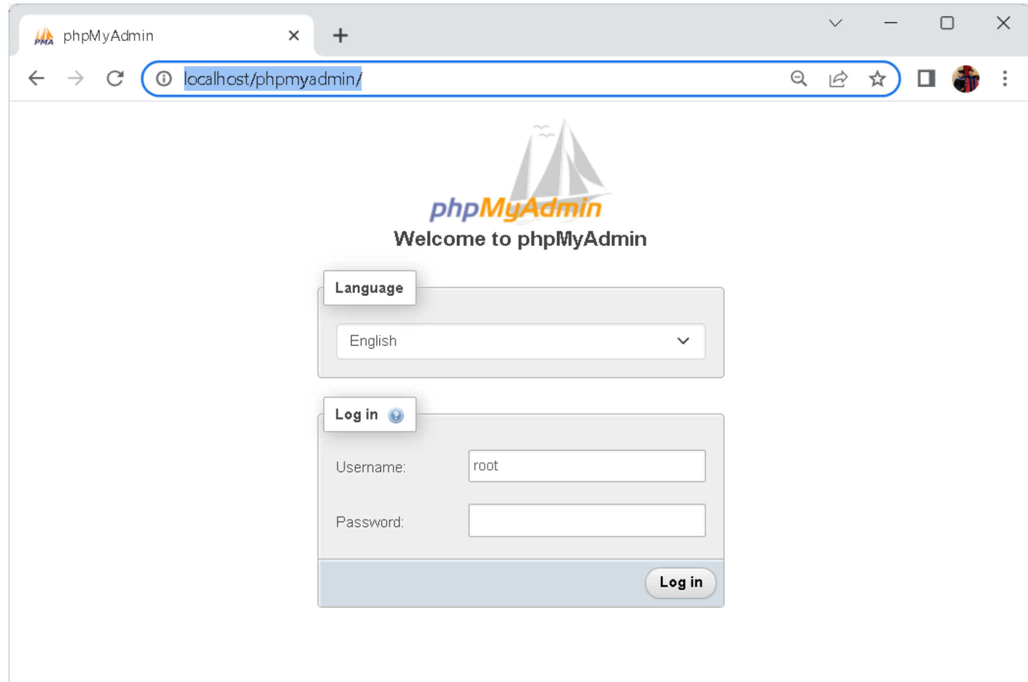
```
Output
Status: active

To                         Action      From
--                         ------      ----
OpenSSH                    ALLOW       Anywhere
Apache                     ALLOW       Anywhere
OpenSSH (v6)               ALLOW       Anywhere (v6)
Apache (v6)                ALLOW       Anywhere (v6)
```

2 – Export & Import SQL Files

EXPORT dan IMPORT SQL files dikenal dengan istilah "**database dump**" atau "**database backup**". Istilah ini merujuk pada **proses export** dan **import data** dari atau ke dalam file SQL yang berisi instruksi SQL untuk *create tabel*, *insert data*, dan mengatur struktur *database*.

“Sign In” *from* Localhost

How *to* Backup Database MySQL?



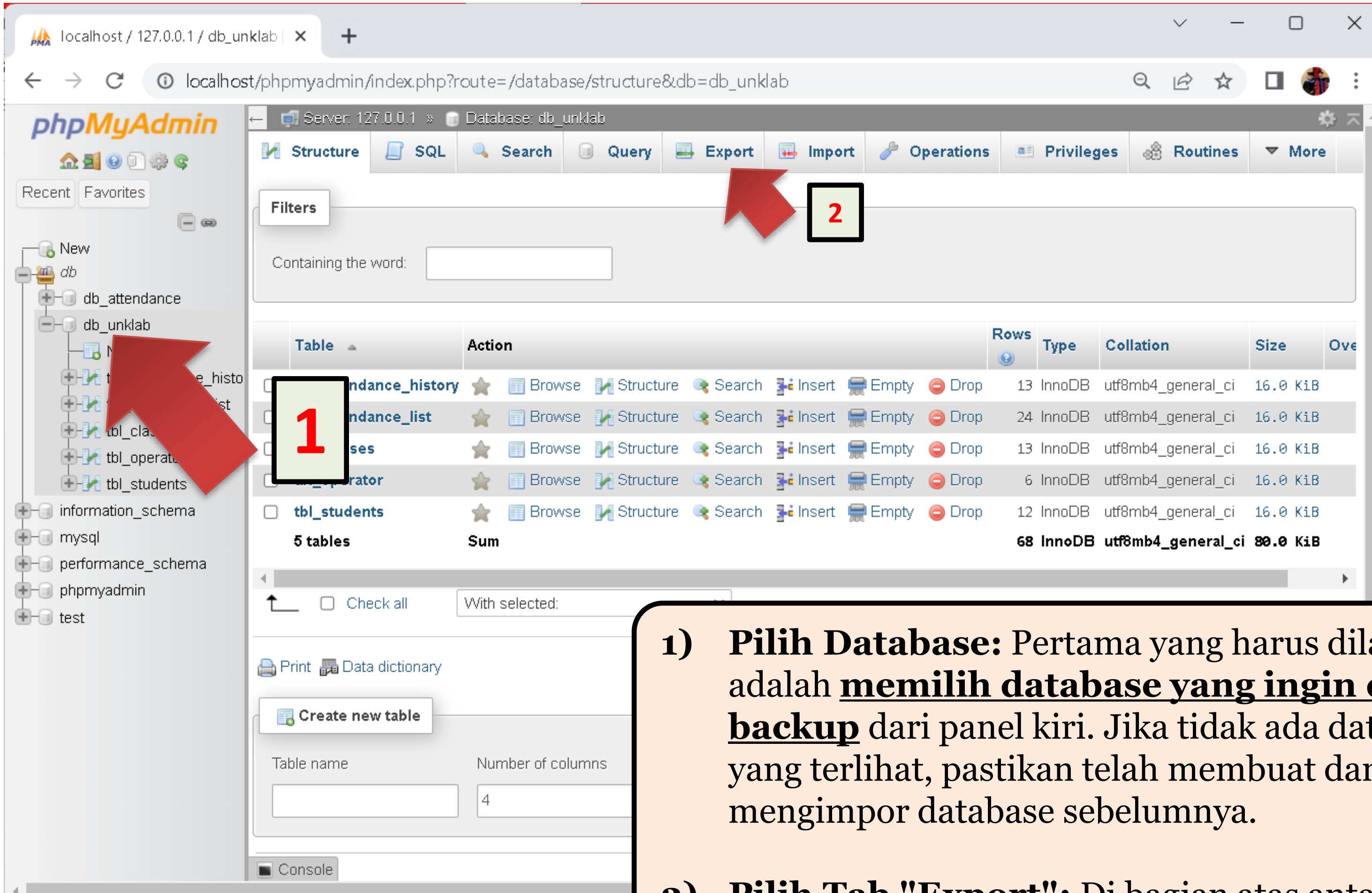
1) **Buka PHPMyAdmin:** Akses halaman PHPMyAdmin melalui browser dengan memasukkan URL yang sesuai. Biasanya, alamat default adalah ***http://localhost/phpmyadmin*** atau sesuai dengan konfigurasi server.

2) **Masuk ke PHPMyAdmin:** Masukkan nama pengguna (username) dan kata sandi (password) MySQL yang valid untuk masuk ke antarmuka PHPMyAdmin.

3) **Pilih Database:** Setelah masuk, **pilih database yang ingin di backup** dari panel kiri. Jika tidak ada database yang terlihat, pastikan telah membuat dan mengimpor database sebelumnya.

4) **Pilih Tab "Export":** Di bagian atas antarmuka PHPMyAdmin, temukan tab "**Export**" dan klik. Ini secara otomatis akan diarahkan ke halaman ekspor database.

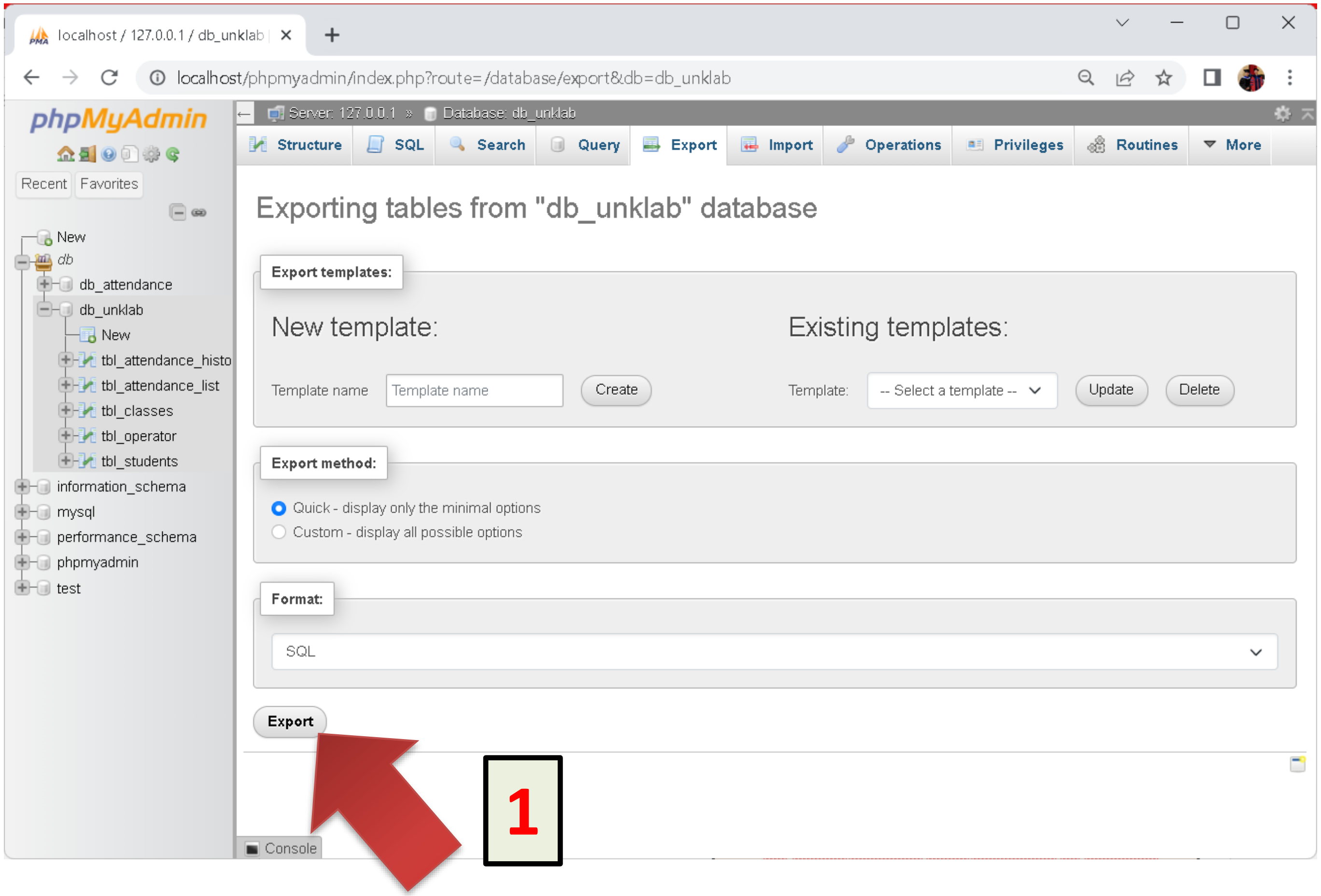
Export MySQL Database



1) **Pilih Database:** Pertama yang harus dilakukan adalah **<u>memilih database yang ingin di backup</u>** dari panel kiri. Jika tidak ada database yang terlihat, pastikan telah membuat dan mengimpor database sebelumnya.

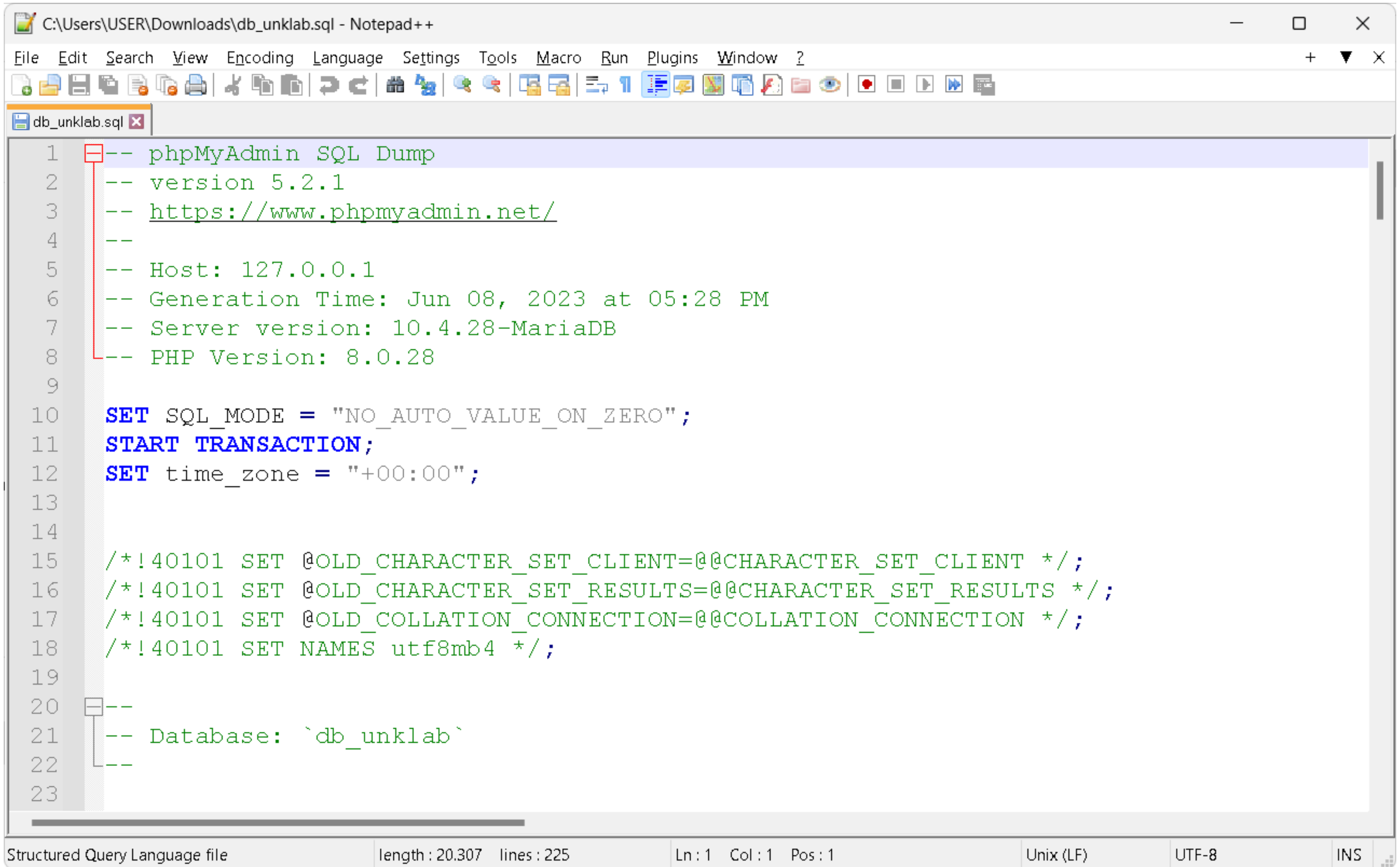
2) **Pilih Tab "Export":** Di bagian atas antarmuka PHPMyAdmin, temukan tab "**Export**" dan klik. Ini secara otomatis akan diarahkan ke halaman ekspor database.

Export MySQL Database



Pilih format *file* yang akan di export, selanjutnya Kalian bisa secara langsung click button “**Export**” dan sistem akan mendownload secara otomatis file dengan nama “***db_unklab.sql***”.

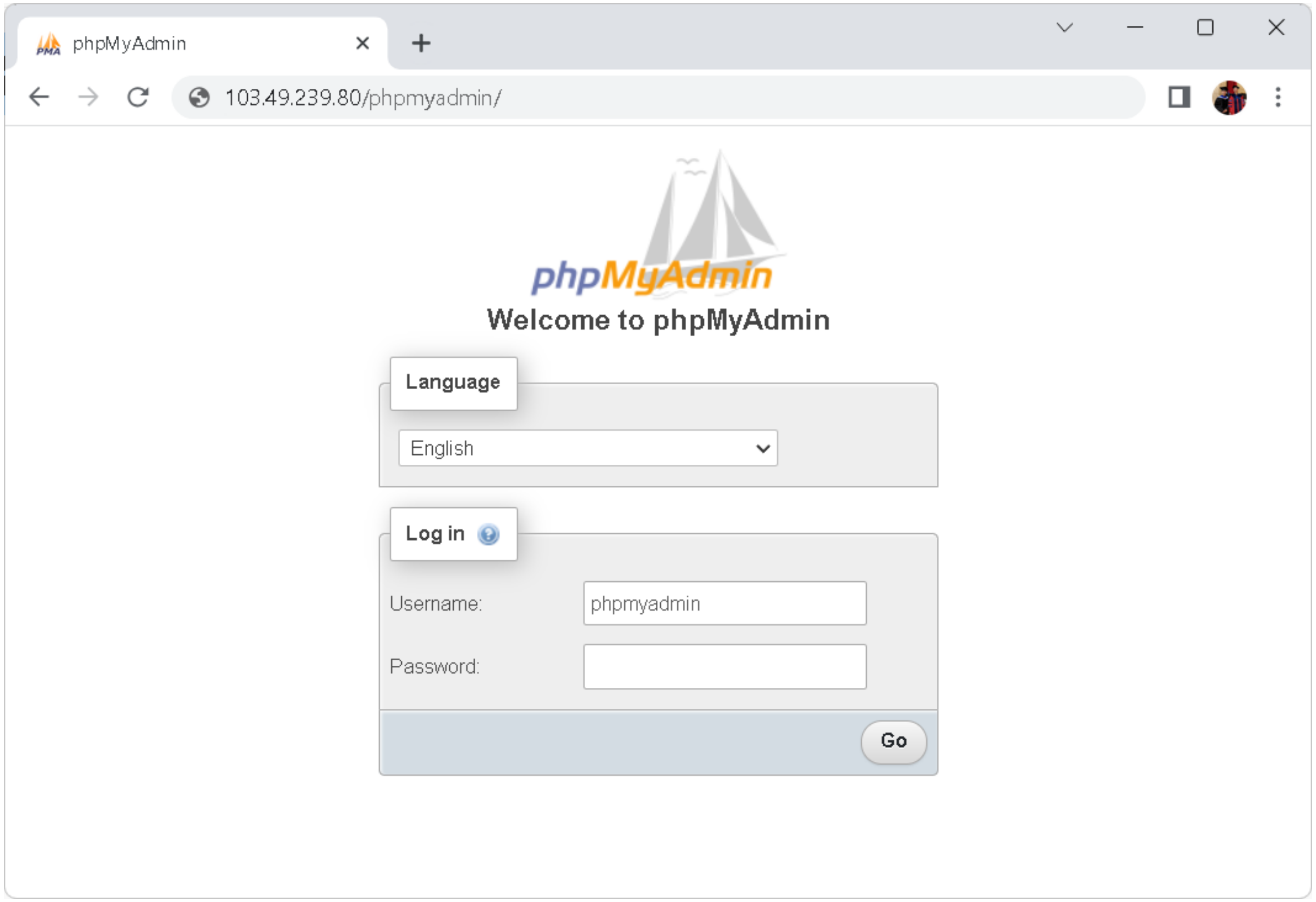
Backup File Downloaded



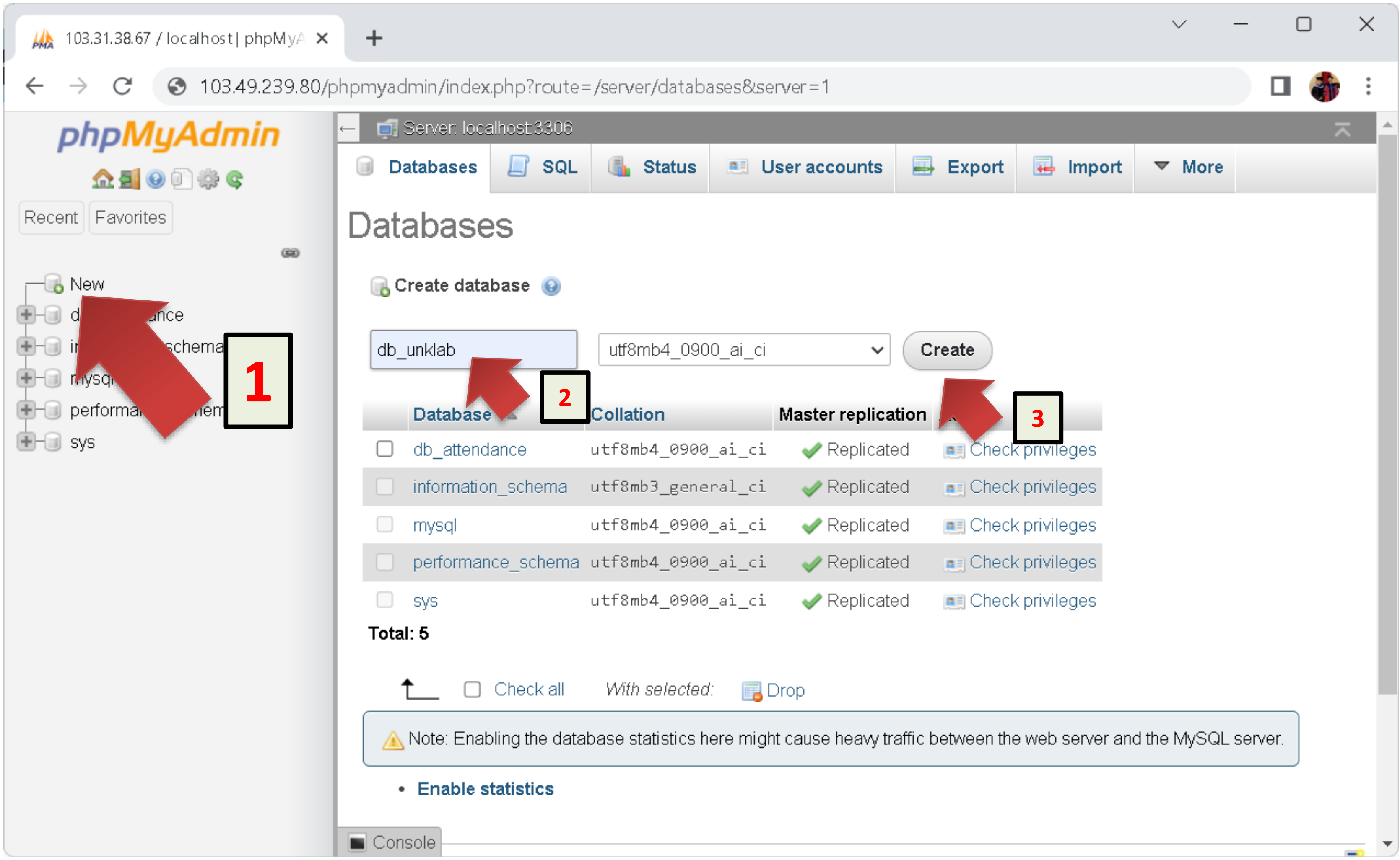

```sql
-- phpMyAdmin SQL Dump
-- version 5.2.1
-- https://www.phpmyadmin.net/
--
-- Host: 127.0.0.1
-- Generation Time: Jun 08, 2023 at 05:28 PM
-- Server version: 10.4.28-MariaDB
-- PHP Version: 8.0.28

SET SQL_MODE = "NO_AUTO_VALUE_ON_ZERO";
START TRANSACTION;
SET time_zone = "+00:00";


/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!40101 SET NAMES utf8mb4 */;

--
-- Database: `db_unklab`
--

```

Bagaimana cara melakukan proses import data ke dalam database MySQL menggunakan backup file SQL yang berisi instruksi SQL ?

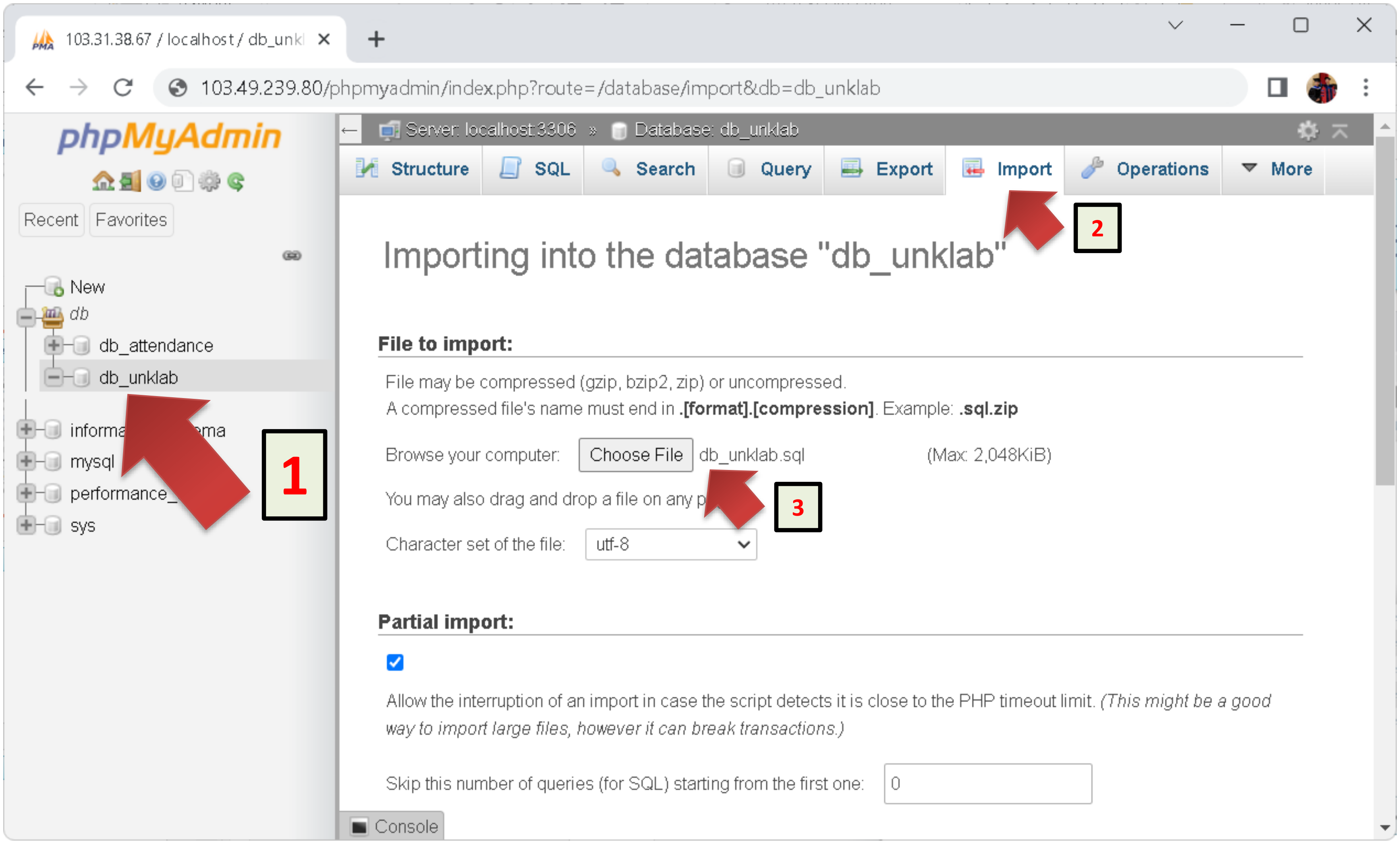
"Sign In" Using Cloud Server



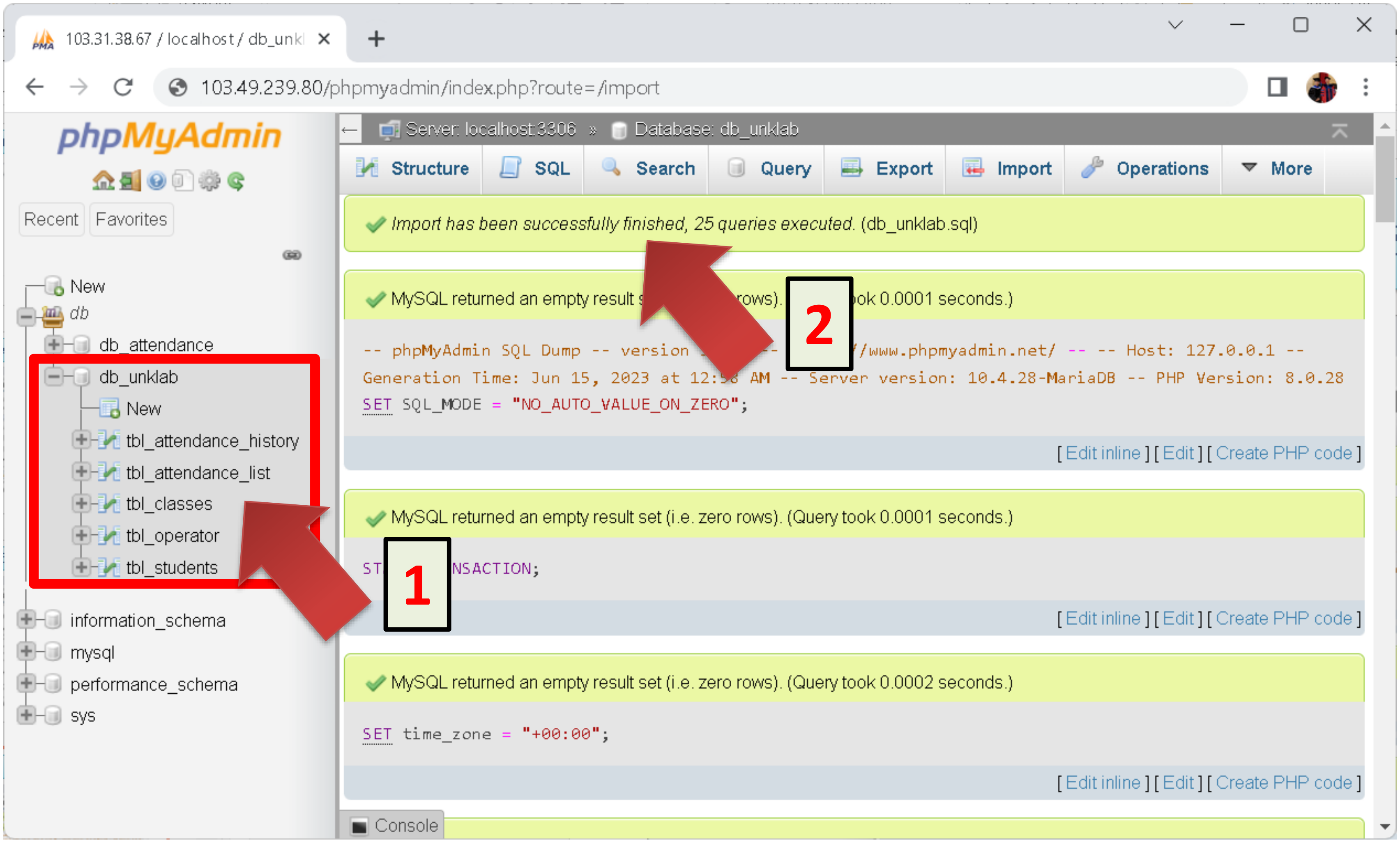
Create New Database Name

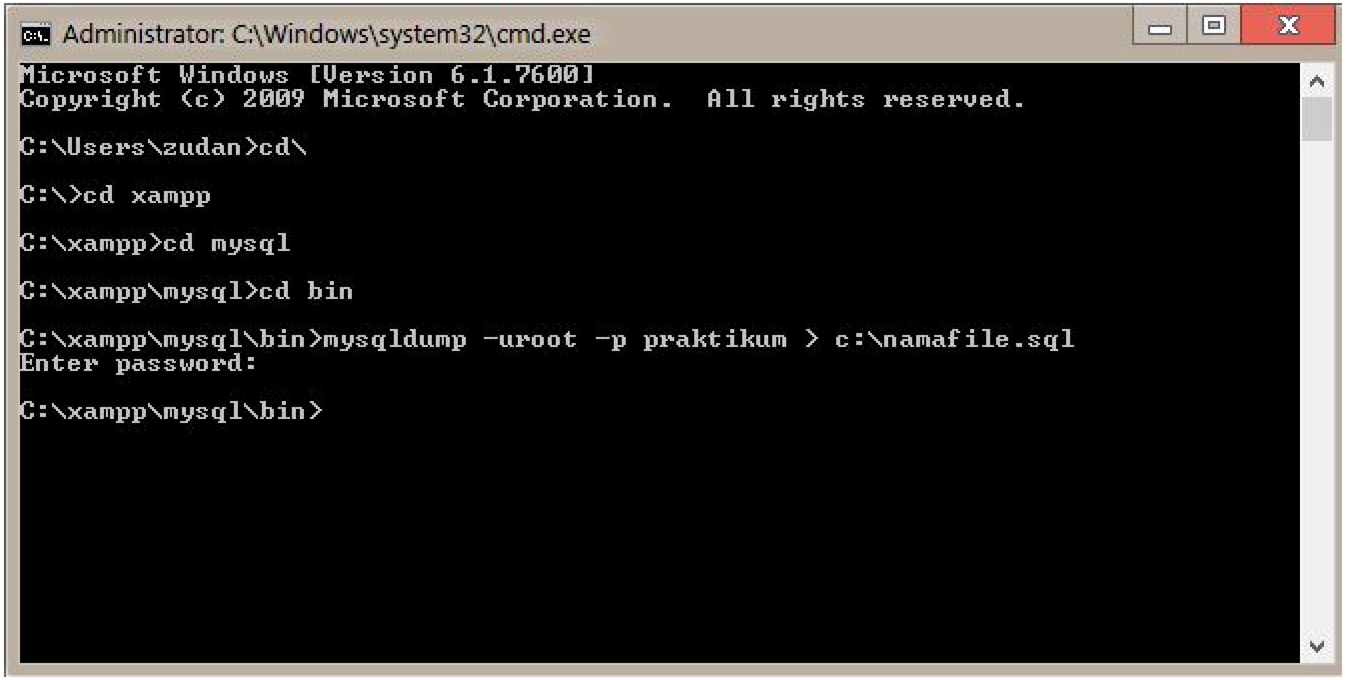


Import SQL File *from* Local PC



Import Successfully Finished



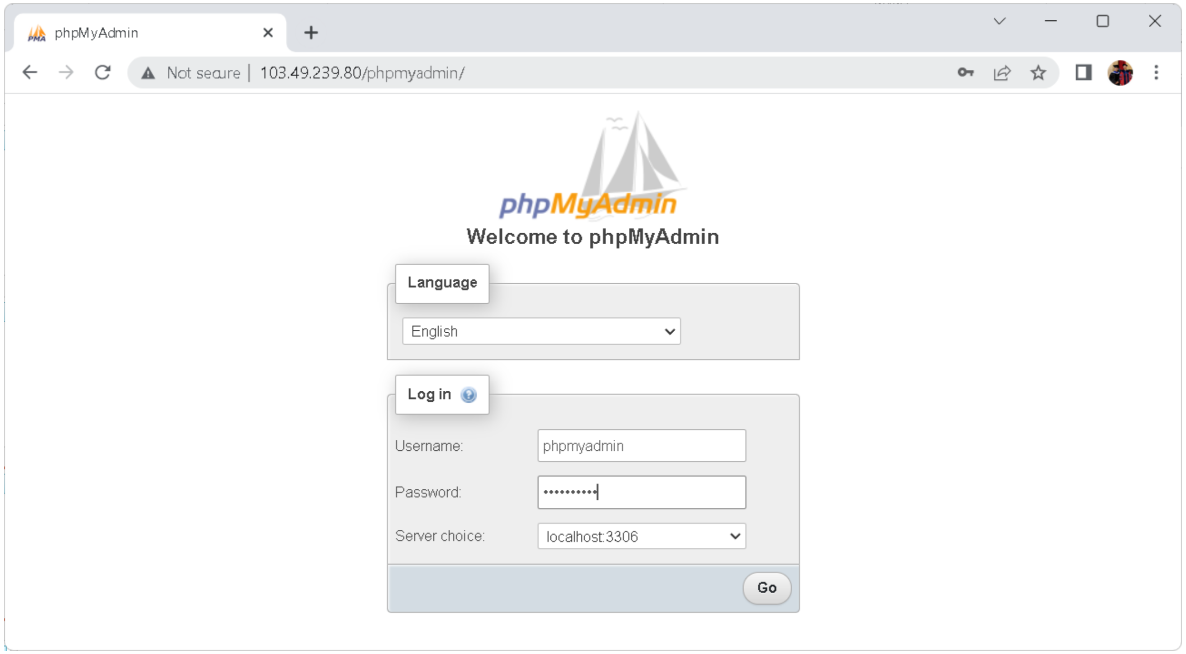


Membuat Dump File:
mysqldump -u username -p nama_database > nama_file_dump.sql

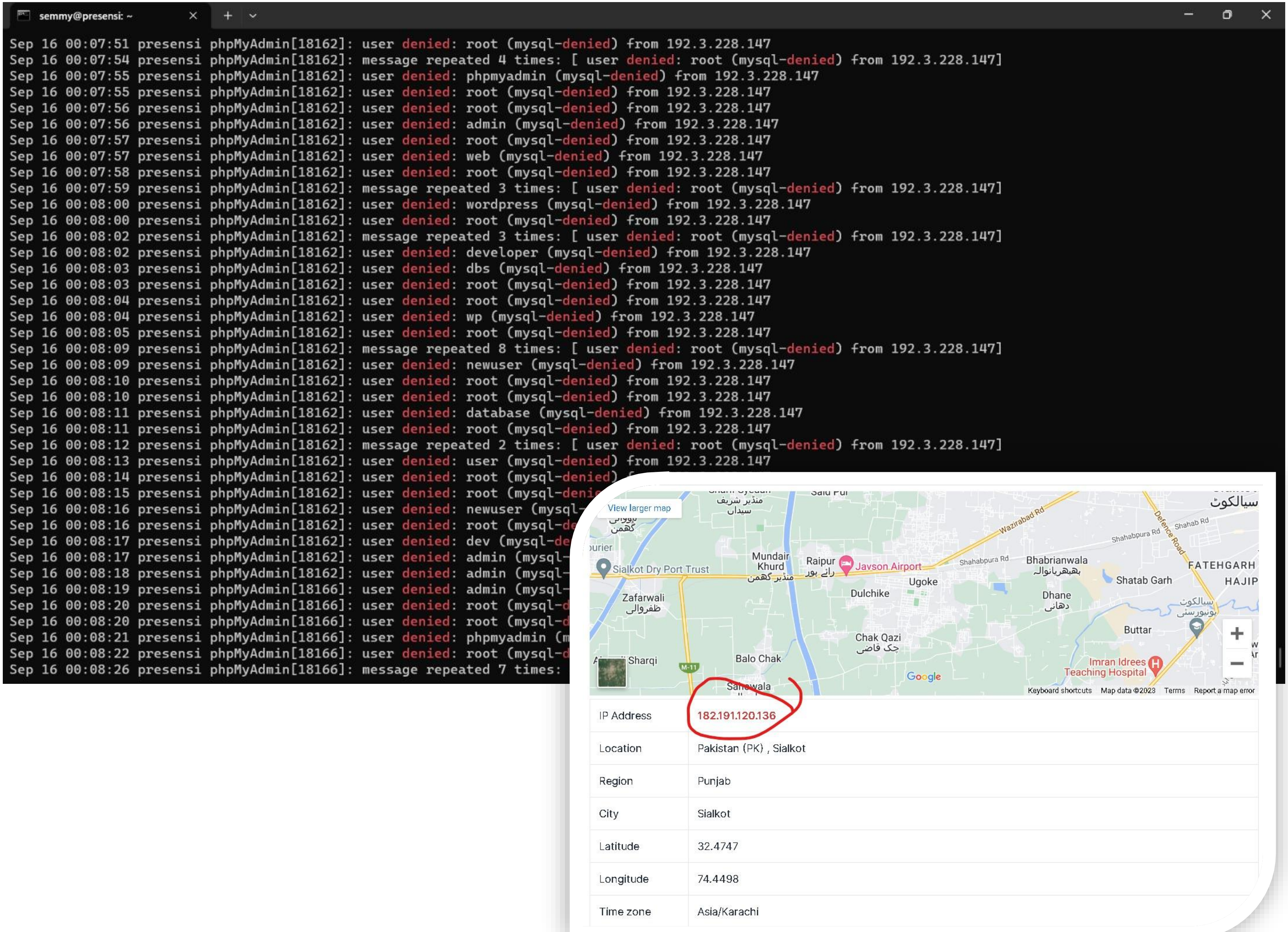
Cara Dump Database Mysql menggunakan Command Prompt (CMD)

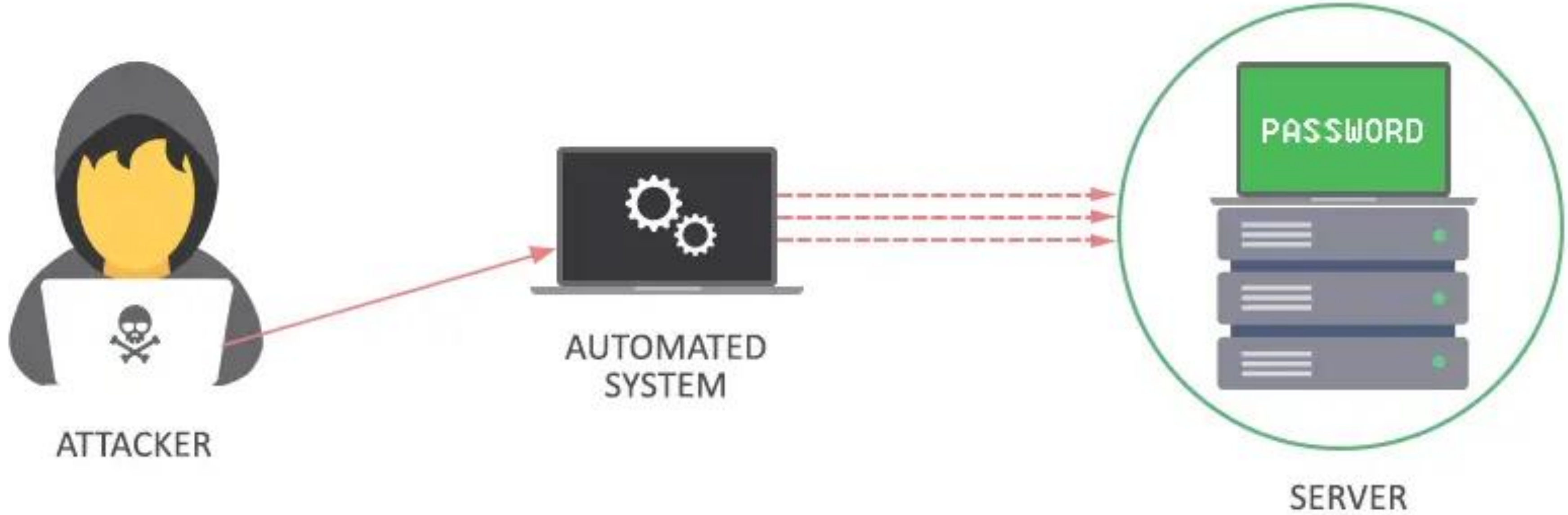
3 – Security PhpMyAdmin *in* Cloud Server?

Apakah **login form (PHPMyAdmin)** ini aman? bagaimana cara agar supaya **login form** ini aman dari unauthorized user?



An ***<u>unauthorized user</u>*** is a *user* of a computer system that does not have authorization to use the system.


semmy@presensi: ~
Sep 16 00:07:51 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:07:54 presensi phpMyAdmin[18162]: message repeated 4 times: [user denied: root (mysql-denied) from 192.3.228.147]
Sep 16 00:07:55 presensi phpMyAdmin[18162]: user denied: phpmyadmin (mysql-denied) from 192.3.228.147
Sep 16 00:07:55 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:07:56 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:07:56 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147
Sep 16 00:07:57 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:07:57 presensi phpMyAdmin[18162]: user denied: web (mysql-denied) from 192.3.228.147
Sep 16 00:07:58 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:07:59 presensi phpMyAdmin[18162]: message repeated 3 times: [user denied: root (mysql-denied) from 192.3.228.147]
Sep 16 00:08:00 presensi phpMyAdmin[18162]: user denied: wordpress (mysql-denied) from 192.3.228.147
Sep 16 00:08:00 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:08:02 presensi phpMyAdmin[18162]: message repeated 3 times: [user denied: root (mysql-denied) from 192.3.228.147]
Sep 16 00:08:02 presensi phpMyAdmin[18162]: user denied: developer (mysql-denied) from 192.3.228.147
Sep 16 00:08:03 presensi phpMyAdmin[18162]: user denied: dbs (mysql-denied) from 192.3.228.147
Sep 16 00:08:03 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:08:04 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:08:04 presensi phpMyAdmin[18162]: user denied: wp (mysql-denied) from 192.3.228.147
Sep 16 00:08:05 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:08:09 presensi phpMyAdmin[18162]: message repeated 8 times: [user denied: root (mysql-denied) from 192.3.228.147]
Sep 16 00:08:09 presensi phpMyAdmin[18162]: user denied: newuser (mysql-denied) from 192.3.228.147
Sep 16 00:08:10 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:08:10 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:08:11 presensi phpMyAdmin[18162]: user denied: database (mysql-denied) from 192.3.228.147
Sep 16 00:08:11 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147
Sep 16 00:08:12 presensi phpMyAdmin[18162]: message repeated 2 times: [user denied: root (mysql-denied) from 192.3.228.147]
Sep 16 00:08:13 presensi phpMyAdmin[18162]: user denied: user (mysql-denied) from 192.3.228.147
Sep 16 00:08:14 presensi phpMyAdmin[18162]: user denied: root (mysql-denied)
Sep 16 00:08:15 presensi phpMyAdmin[18162]: user denied: root (mysql-deni
Sep 16 00:08:16 presensi phpMyAdmin[18162]: user denied: newuser (mysql-
Sep 16 00:08:16 presensi phpMyAdmin[18162]: user denied: root (mysql-de
Sep 16 00:08:17 presensi phpMyAdmin[18162]: user denied: dev (mysql-de
Sep 16 00:08:17 presensi phpMyAdmin[18162]: user denied: admin (mysql-
Sep 16 00:08:18 presensi phpMyAdmin[18162]: user denied: admin (mysql-
Sep 16 00:08:19 presensi phpMyAdmin[18166]: user denied: admin (mysql-
Sep 16 00:08:20 presensi phpMyAdmin[18166]: user denied: root (mysql-d
Sep 16 00:08:20 presensi phpMyAdmin[18166]: user denied: root (mysql-d
Sep 16 00:08:21 presensi phpMyAdmin[18166]: user denied: phpmyadmin (m
Sep 16 00:08:22 presensi phpMyAdmin[18166]: user denied: root (mysql-d
Sep 16 00:08:26 presensi phpMyAdmin[18166]: message repeated 7 times:
View larger map
Sialkot Dry Port Trust
Mundair Khurd
Raipur
Javson Airport
Ugoke
Dulchike
Zafarwali
Chak Qazi
Balo Chak
Sharqi
Bhabrianwala
Shatab Garh
Dhane
Buttar
FATEHGARH
HAJIP
Imran Idrees Teaching Hospital
Wazirabad Rd
Shahabpura Rd
Defence Road
Google
Keyboard shortcuts
Map data ©2023
Terms
Report a map error
IP Address 182.191.120.136
Location Pakistan (PK) , Sialkot
Region Punjab
City Sialkot
Latitude 32.4747
Longitude 74.4498
Time zone Asia/Karachi



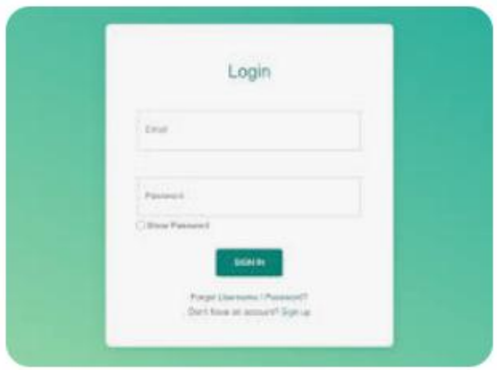
Brute Force Attack adalah metode untuk melakukan membobol sebuah akun dengan cara menebak *username* dan bahkan *password* yang digunakan oleh users (*pengguna*) tersebut. Metode ini akan dilakukan dengan berbagai *experiment* melalui *trial* and *error* hingga akhirnya berhasil untuk masuk ke dalam *account* yang ditargetkan.

Exercise #1 *for* Students

(Automated Backup Database)

Setiap kelompok diminta untuk membuat landing page, login form dan dashboard sebuah sistem yang focus pada sektor-sektor industri penting (pertanian, perikanan, tambang, dll) di Sulawesi Utara.

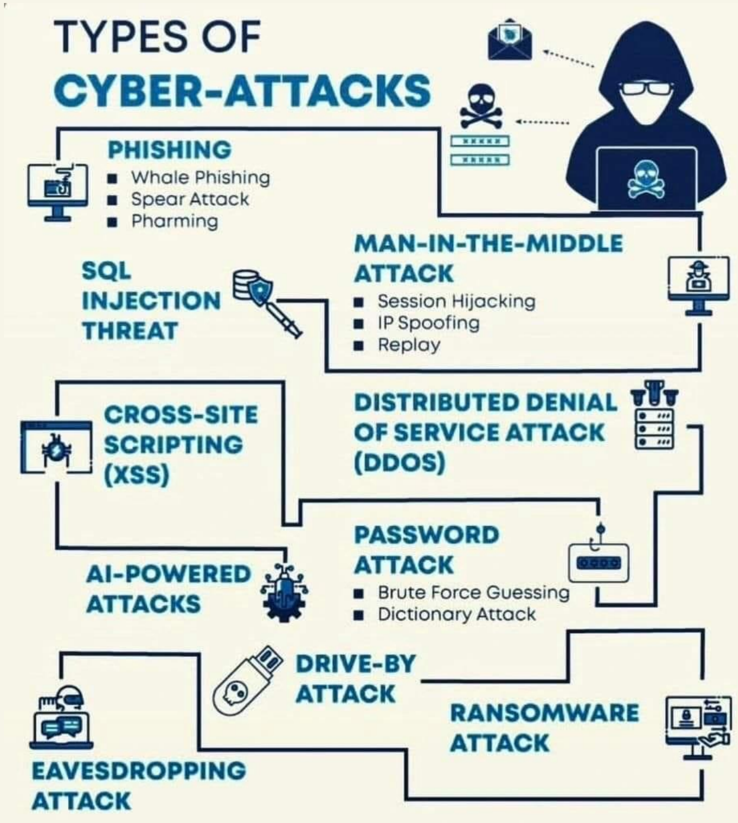
Landing Page
Halaman web yang dimaksudkan untuk mengubah pengunjung web menjadi pelanggan atau prospek. Halaman ini biasanya memiliki tampilan sederhana, fokus pada satu tindakan, dan elemen desain yang menarik untuk meningkatkan tingkat konversi.



Login Form
Pintu masuk keamanan (*autentikasi*) berbasis website atau aplikasi yang mengharuskan pengguna memasukkan kredensial, biasanya username/email dan password, untuk memverifikasi identitas dan mendapatkan akses hak guna.

Dashboard
Antarmuka visual yang menyajikan data, metrik, dan Key Performance Indicator (KPI) penting secara real-time atau berkala dalam satu tampilan.

- ✓ Phishing (digital fraud) means an attacker pretends to be an official or trusted party to steal data. For example, a fake email pretending to be from Google asks the victim to log into their account. Whale Phishing targets high-ranking officials, Spear Phishing is tailored for a specific individual, and Pharming redirects victims to fake websites without their awareness.
- ✓ SQL Injection means an attacker inserts malicious code into a website form. For example, a login form is exploited to gain unauthorized access to a user database.
- ✓ Man-in-the-Middle Attack happens when an attacker intercepts communication between two parties. For example, public Wi-Fi is used to steal login credentials.
- ✓ Session Hijacking occurs when an attacker steals an active login session token; for example, a social media account is taken over without needing the password.
- ✓ IP Spoofing means an IP address is forged to appear as if it comes from a trusted source; for example, a server receives fake traffic that appears to come from inside the office network.
- ✓ Replay Attack happens when valid communication data is recorded and resent; for example, an old transaction code is reused to gain access.
- ✓ Cross-Site Scripting (XSS) is when malicious scripts are inserted into a website; for example, a website comment contains code that steals login cookies.
- ✓ DDoS (Distributed Denial of Service) is when many devices attack a server at the same time; for example, an online store website goes down because of massive fake traffic.
- ✓ Password Attack means passwords are stolen or guessed; for example, an email account is hacked because the password is weak.
- ✓ Brute Force Attack means the system tries many password combinations; for example, a program attempts thousands of login combinations.
- ✓ Dictionary Attack means passwords are guessed using common word lists; for example, using "123456" or "password".
- ✓ AI-Powered Attacks use artificial intelligence to make attacks more advanced; for example, a fake chatbot imitates a manager's speaking style.
- ✓ Drive-By Attack happens when malware enters a device simply by visiting a malicious website; for example, an automatic download starts from an unsafe site.
- ✓ Ransomware locks files and demands payment; for example, all office files are encrypted and the victim is asked to pay a ransom.
- ✓ Eavesdropping means secretly listening to digital communication; for example, chat data is stolen through an unsecured network.



Sebutkan tipe-tipe serangan keamanan yang bisa mengancam **database server**?

Bagaimana cara anda untuk menangani tipe serangan seperti ini?

Buatlah demo **proses otomatis untuk melakukan backup data database MySQL pada server Ubuntu 22.04 menggunakan Jenkins**.

Class Practical Terakhir:

Buat report lengkap *step by step* akan dikumpul di GCR dan next meeting students akan diminta untuk melakukan demo Automated backup database di dalam class !!!

Perhatikan contoh *shell script file* dibawah ini yang akan digunakan nanti di Jenkins.

```bash
#!/bin/bash

# MySQL Database Credentials
DB_USER="semmy"
DB_PASS="12345edcba"
DB_NAME="db_university"

# Path to store dump file
BACKUP_DIR="/home/semmy/backup"

# Dump file name
DUMP_FILE="$BACKUP_DIR/dump_db_university.sql"

# Create backup directory
mkdir -p $BACKUP_DIR

# Create dump file
mysqldump -u $DB_USER -p$DB_PASS $DB_NAME > $DUMP_FILE
```

How to create shell script dump file for Jenkins?

1) Create a shell script file in Jenkins that will run the mysqldump command.
   Example: *backup_mysql.sh*

2) Edit shell script file and add the mysqldump command to create a dump file of the desired database. See the picture below.

3) Change values of DB_USER, DB_PASS, DB_NAME, and BACKUP_DIR variables according to MySQL server settings and the location where you want to save dump file.

4) Grant execution privileges to shell script file so it can be executed.
   > *chmod +x backup_mysql.sh*

5) Edit Jenkins job and run the MySQL backup. Don't forget to dd shell command to run the script:
   > *sh /path/to/backup_mysql.sh*

Exercise #2 *for* Students

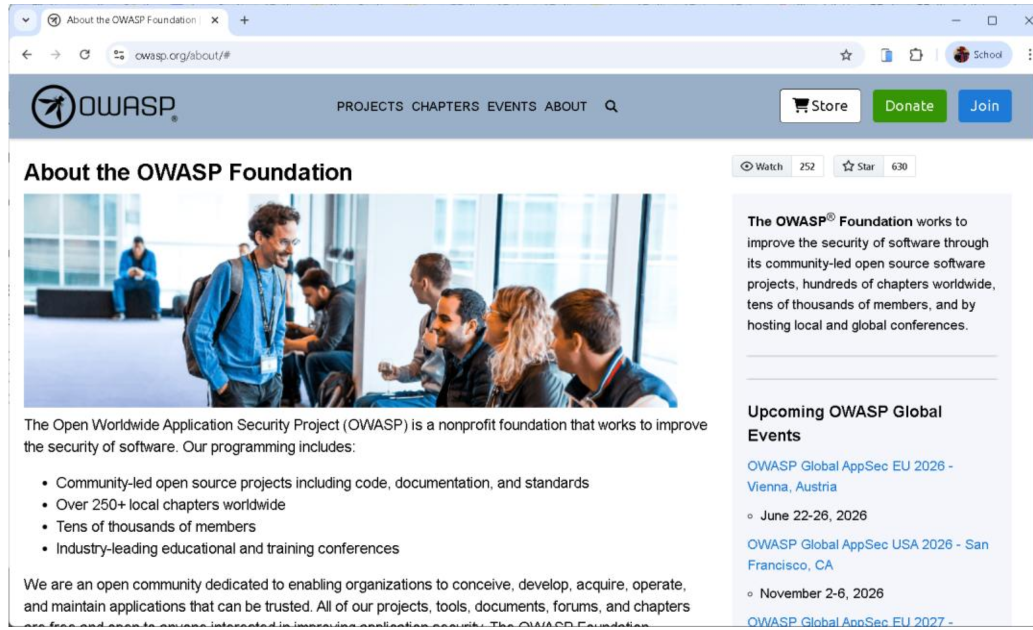
OWASP (Open Web Application Security Project)

What is OWASP?

OWASP security audit is a security assessment of an application based on standards and guidance from **OWASP (Open Web Application Security Project)**. OWASP is a global, non-profit organization that provides free, community-driven security resources for web and application security.

What does it usually check?
Most OWASP audits are based on **OWASP Top 10**, such as:

1) **Broken Access Control**
2) **Cryptographic Failures**
3) **Injection attacks** (SQL, Command, XSS)
4) **Insecure Design**
5) **Security Misconfiguration**
6) **Vulnerable & Outdated Components**
7) **Identification & Authentication Failures**
8) **Software & Data Integrity Failures**
9) **Security Logging & Monitoring Failures**
10) **Server-Side Request Forgery (SSRF)**



OWASP Security Audit

Nama Sistem	
Unit Pengelola	

Teknologi	`Programming Language :`
	`Back-end Framework :`
	`Front-end Framework :`
	`Database :`
	`Web Server :`
	`Operating System (OS) :`
	`Arsitektur (Web-based) :`

URL Sistem	
Auditor	
Tanggal Audit	
Firewall Server	☐ `Block Port 22 (SSH)` ☐ `Allow Port 80 (HTTP)` ☐ `Allow Port 443 (HTTPS)` ☐ `Block Port 3306 (Database)` ☐ `Block Port 25/587/465 (SMTP)`
Link Bukti Audit	

A. Authentication & Session Management (10%)

(OWASP A07 – Identification and Authentication Failures)

No	Item Pemeriksaan	Yes	No	Catatan
A1	(2%) Sistem tidak menggunakan session ID di URL	☐	☐	
A2	(1%) Session disimpan hanya melalui cookie	☐	☐	
A3	(1%) Cookie session menggunakan HttpOnly	☐	☐	
A4	(1%) Cookie session menggunakan Secure (HTTPS)	☐	☐	
A5	(1%) Cookie session menggunakan SameSite	☐	☐	
A6	(1%) Session ID diregenerasi setelah login	☐	☐	
A7	(1%) Session timeout diterapkan (≤ 30 menit)	☐	☐	
A8	(1%) Session otomatis logout saat *idle* (*tidak ada aktivitas dalam suatu periode waktu tertentu*)	☐	☐	
A9	(1%) Session di *destroy* (dihancurkan) saat logout	☐	☐	

Developers can learn from the mistakes of other organizations.

OWASP Security Audit

top10proactive.owasp.org/archive/2024/

OWASP Top 10 Proactive Controls

Search

Future Top 10 (WIP) Top 10 2024 Top 10 2018 Top 10 2016 Top 10 2014

Top 10 2024

In the News

How to contribute?

About OWASP

Related OWASP Projects

Top 10 Proactive Controls:

C1: Implement Access Control

C2: Use Cryptography to Protect Data

C3: Validate all Input & Handle Exceptions

C4: Address Security from the Start

C5: Secure By Default Configurations

C6: Keep your Components Secure

C7: Secure Digital Identities

About this Project

Insecure software is undermining our financial, healthcare, defense, energy, and other critical infrastructure worldwide. As our digital, global infrastructure gets increasingly complex and interconnected, the difficulty of achieving application security increases exponentially. We can no longer afford to tolerate relatively simple security problems.

Aim & Objective

The goal of the **OWASP Top 10 Proactive Controls project** is to raise awareness about application security by describing the most important areas of concern that software developers must be aware of. We encourage you to use the OWASP Proactive Controls to get your developers started with application security. Developers can learn from the mistakes of other organizations. We hope that the OWASP Proactive Controls is useful to your efforts in building secure software.

OWASP Audit *vs* Penetration Test

OWASP Standards
(Pemeriksaan keamanan sistem berdasarkan standar)

Risk & Compliance

Attacker Mindset
(Pentest adalah simulasi serangan nyata)

Attack Paths

Lakukan OWASP security audit untuk melihat kepatuhan dalam standard keamanan aplikasi web.

END PRESENTATION

Thank you for your attention

Instructor: S – W – T